

게임엔진 기초

Unity / Godot 사용해 보기

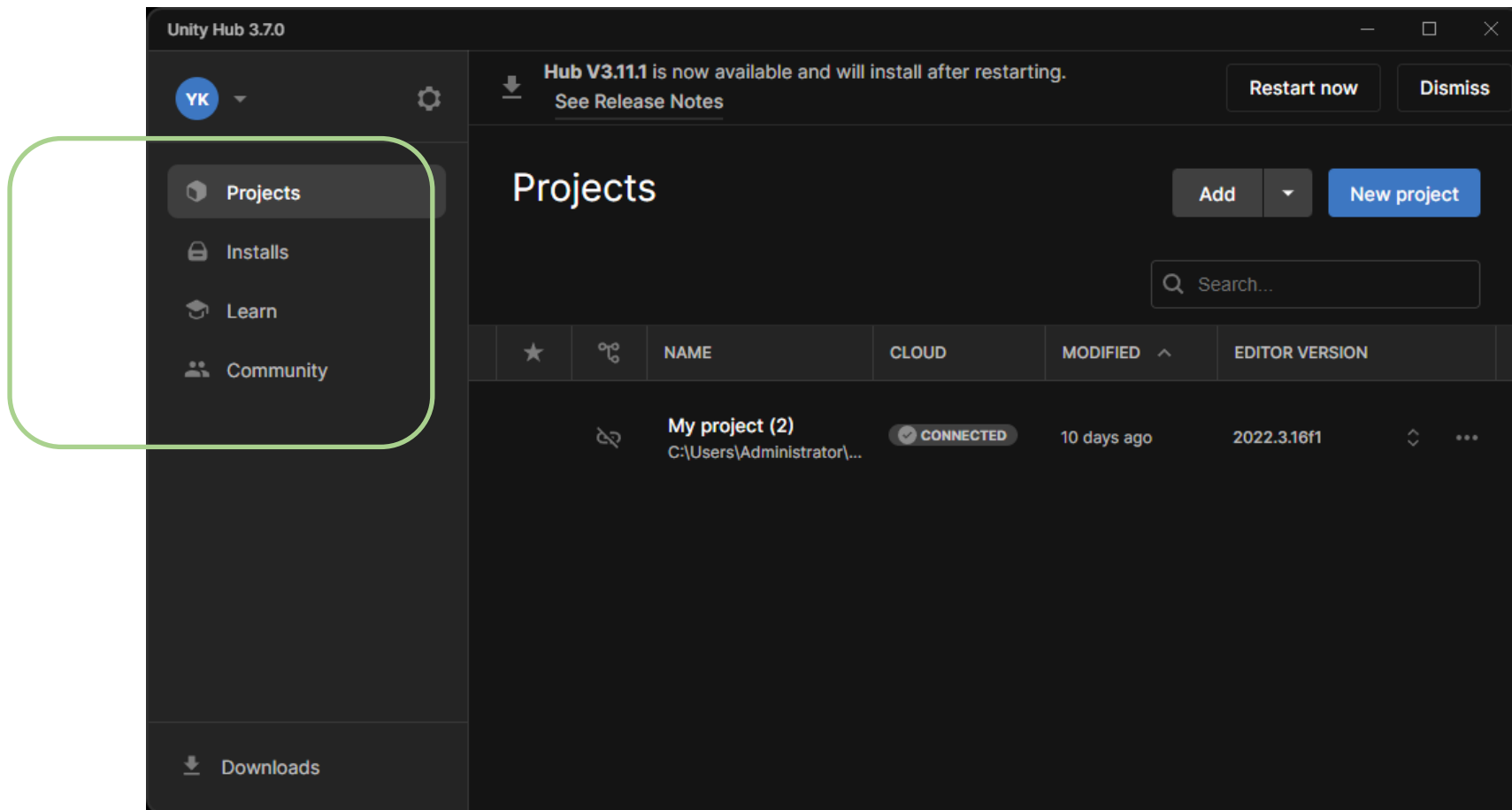
강영민

동명대학교 게임그래픽학과

Unity 를 이용한 간단한 2D 게임 환경 구축

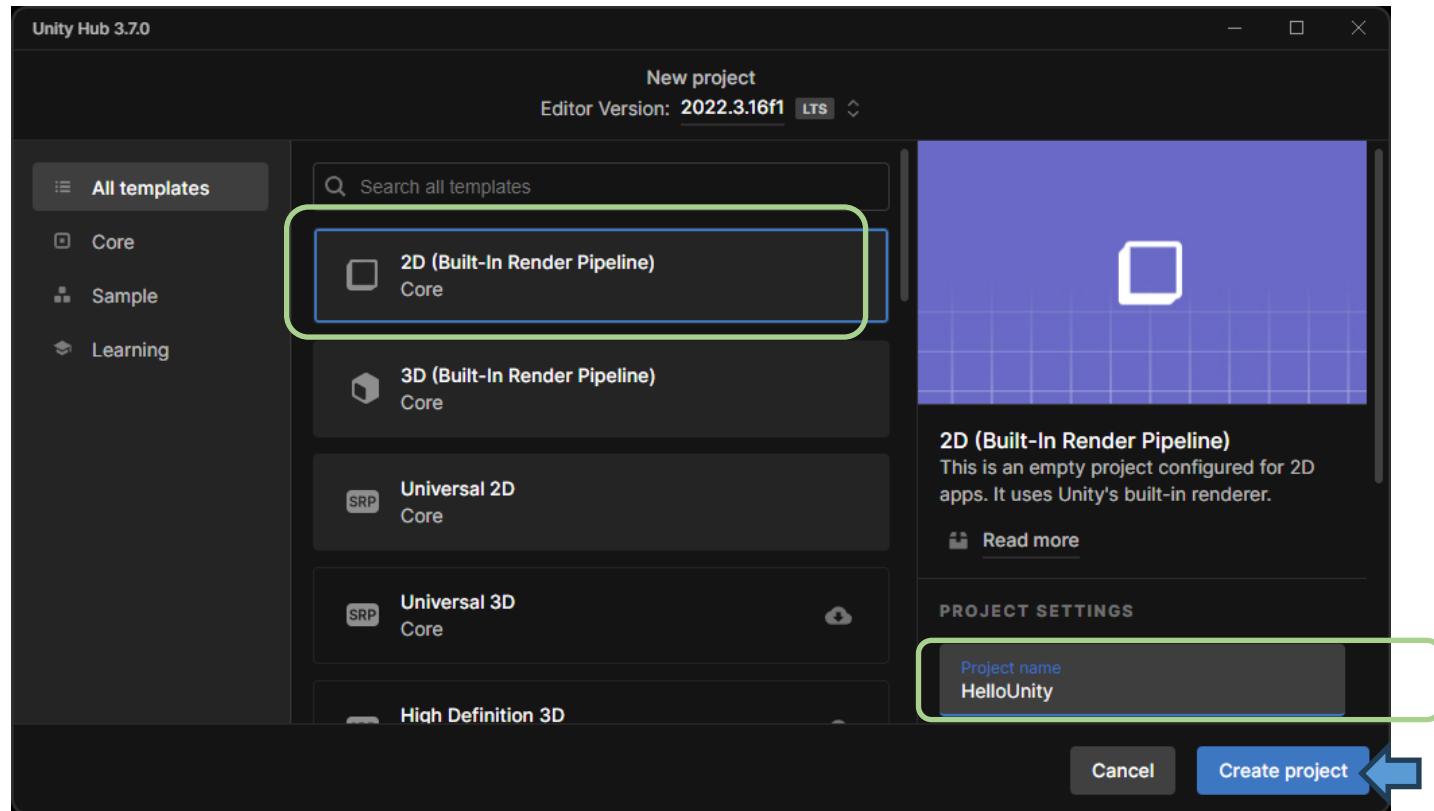
설치한 Unity Hub를 실행해 보자

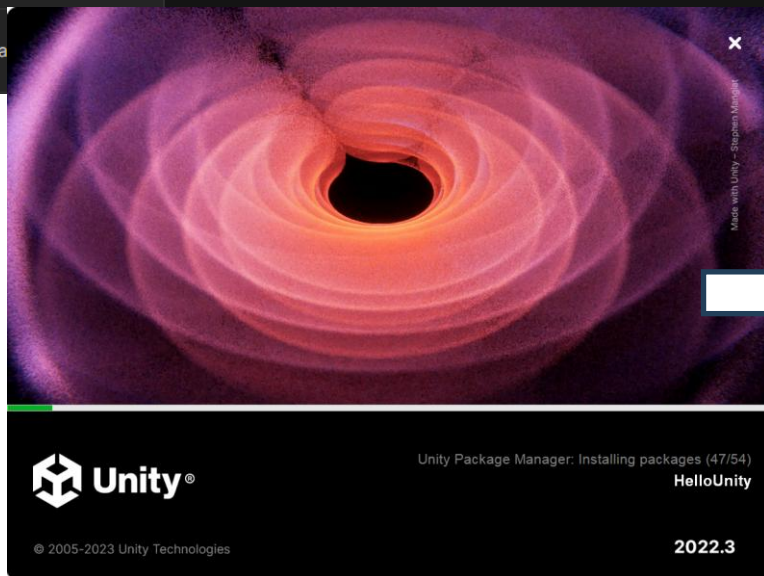
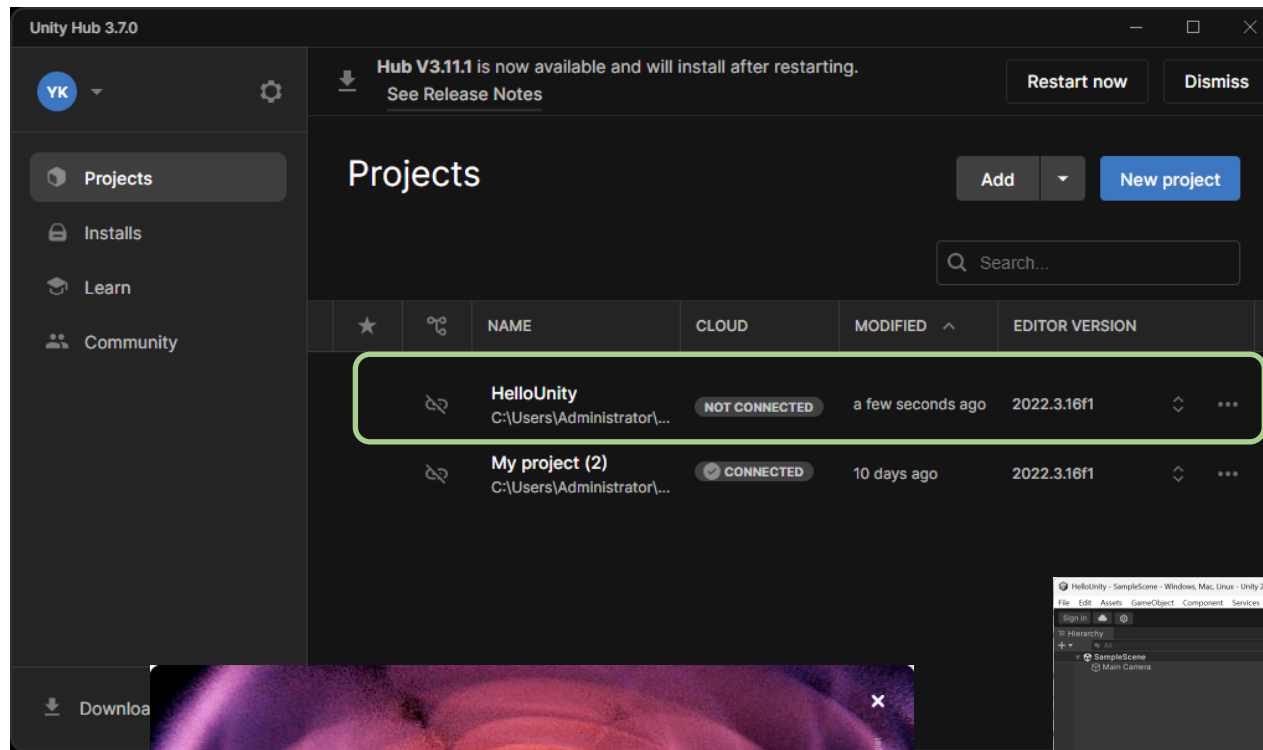
- 네 개의 탭이 탭 메뉴가 존재



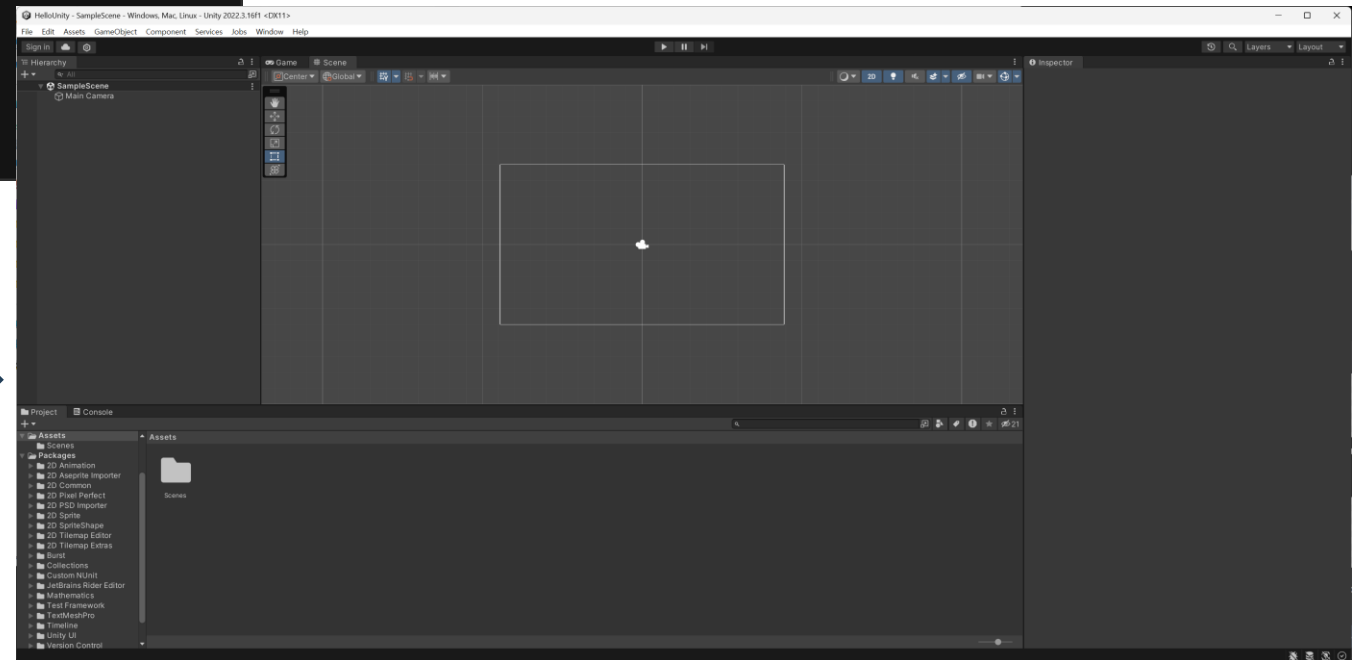
2D 게임프로젝트를 생성해 보자

- 네 개의 탭이 탭 메뉴가 존재



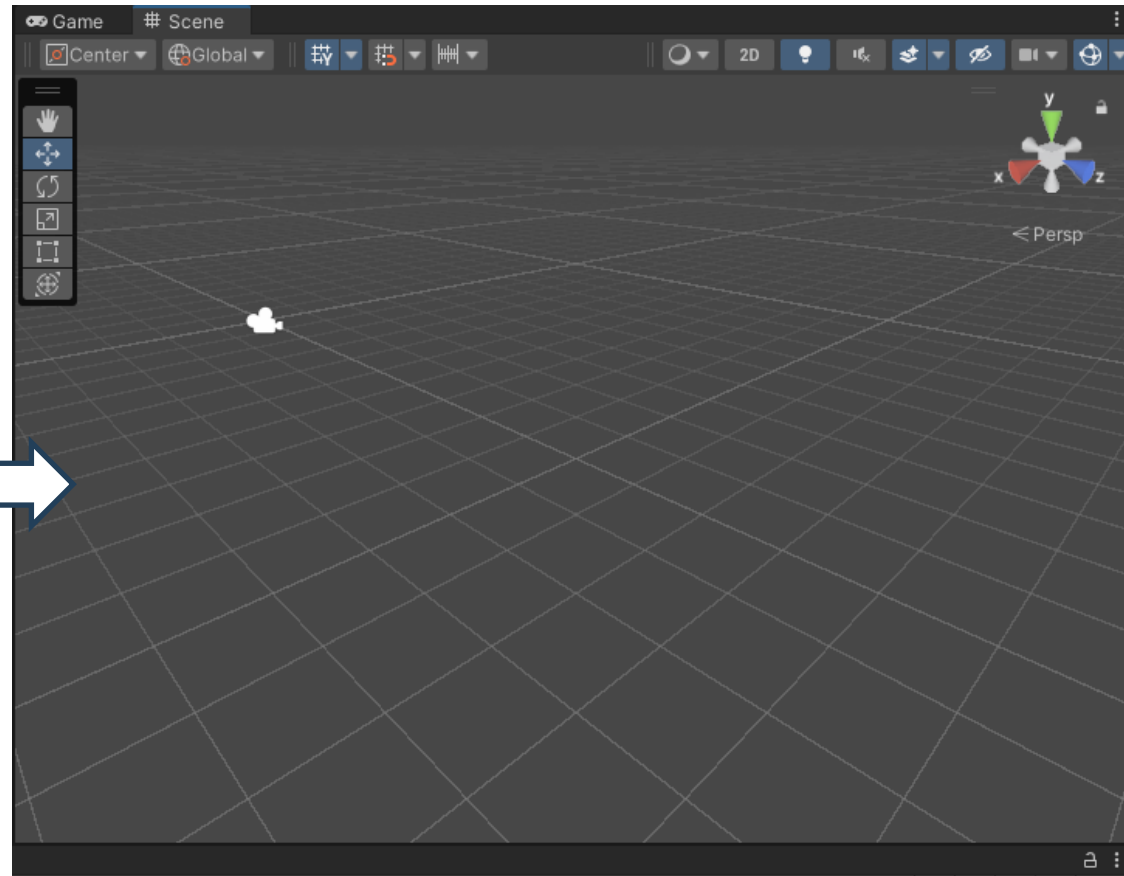
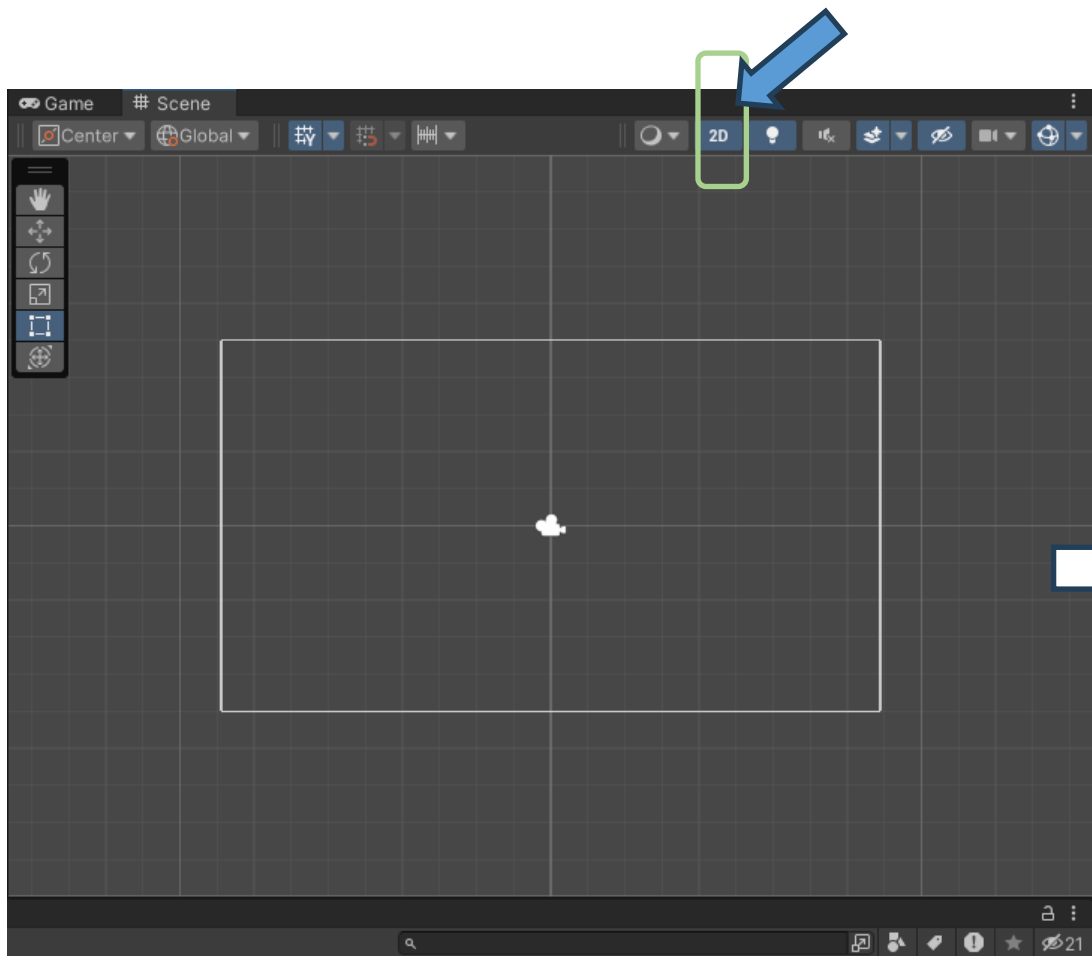


유니티 엔진 실행됨

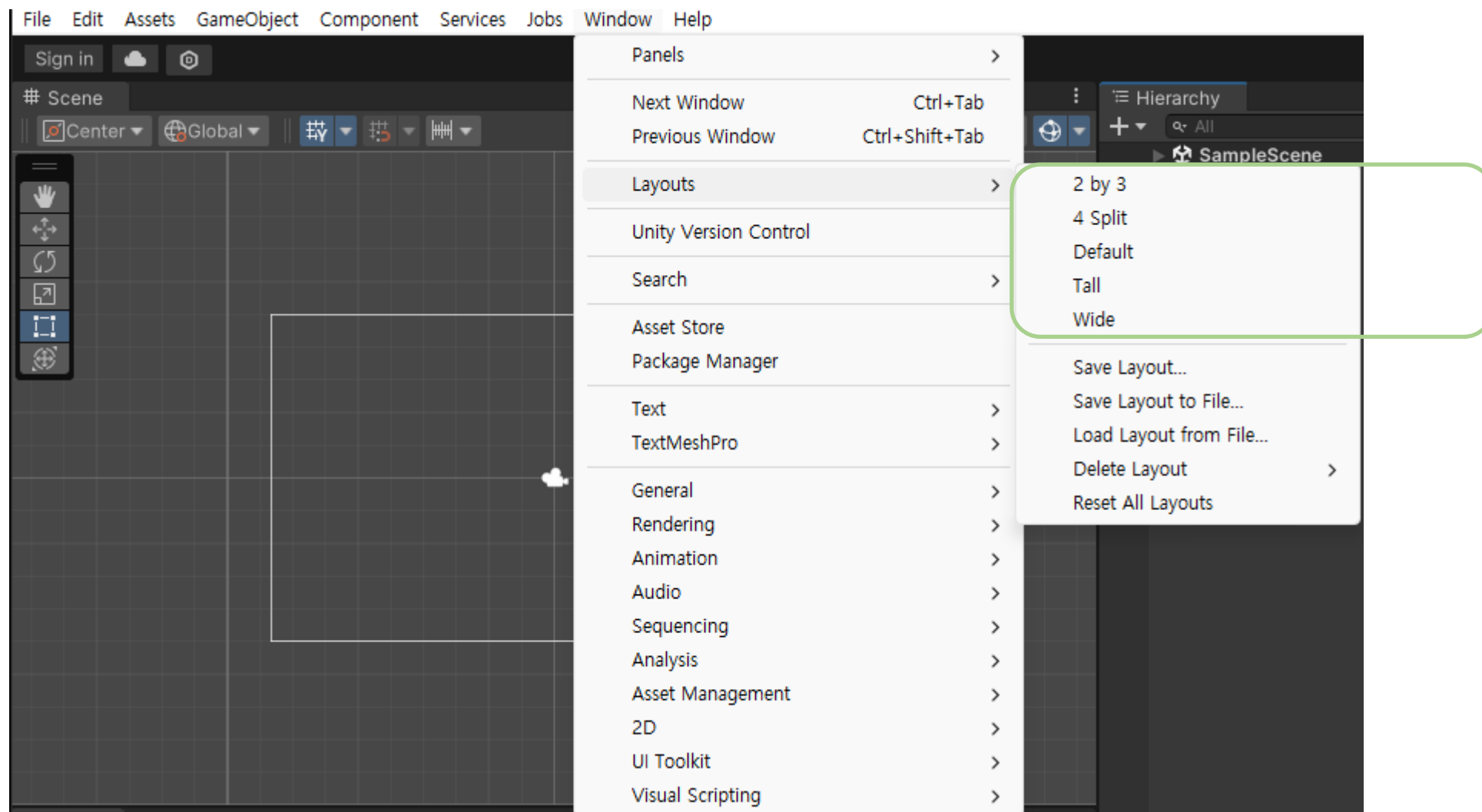


유니티 엔진 실행되고 프로젝트가 로딩된 상태

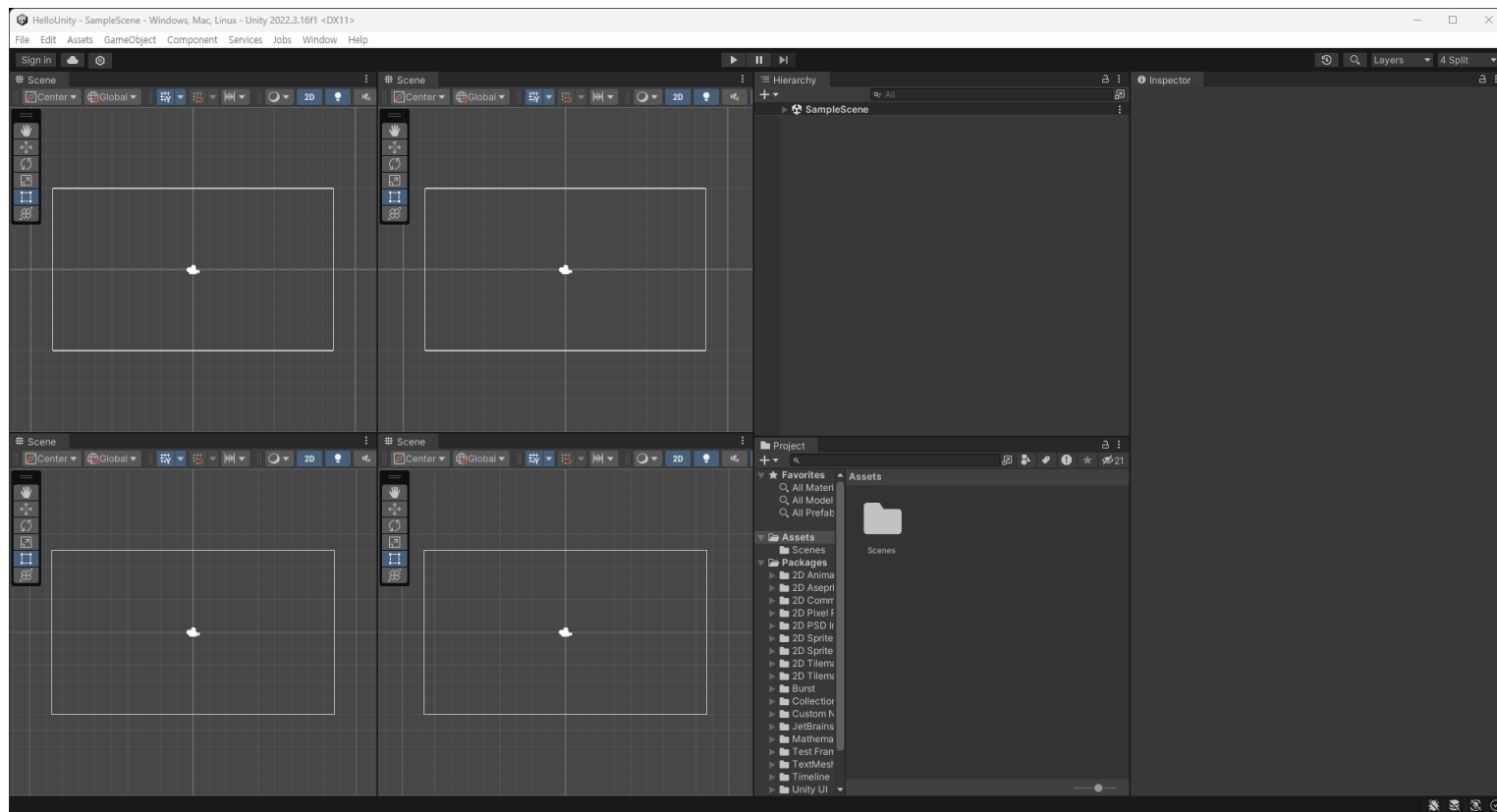
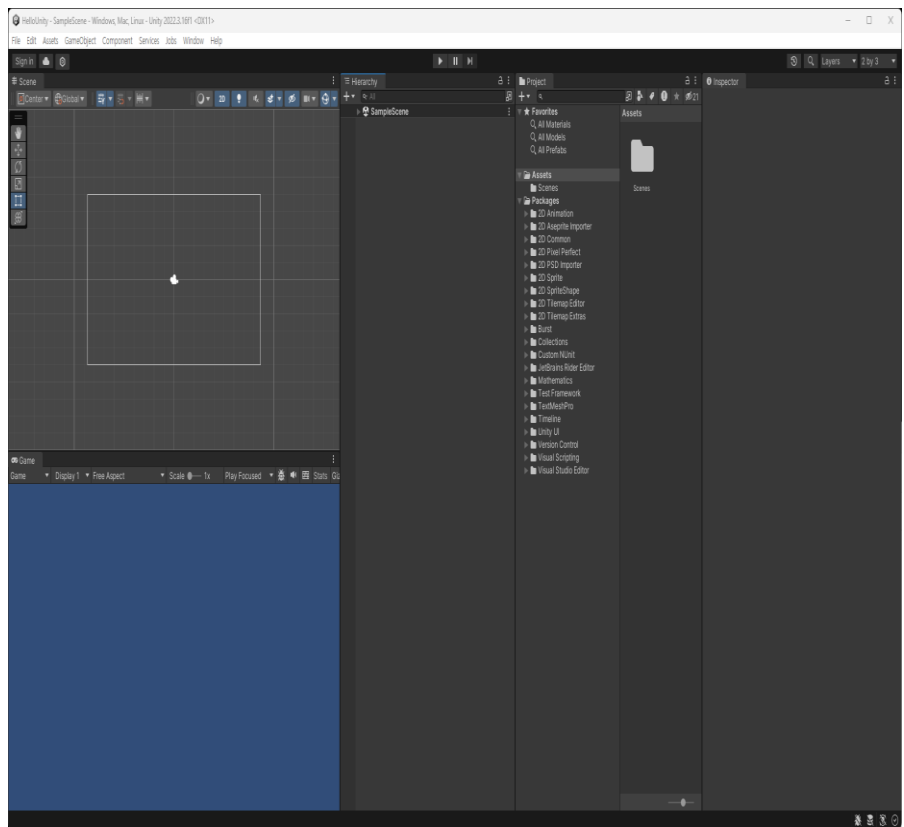
- 2D 공간은 3D 공간의 일부



다양한 윈도우 레이아웃 테스트

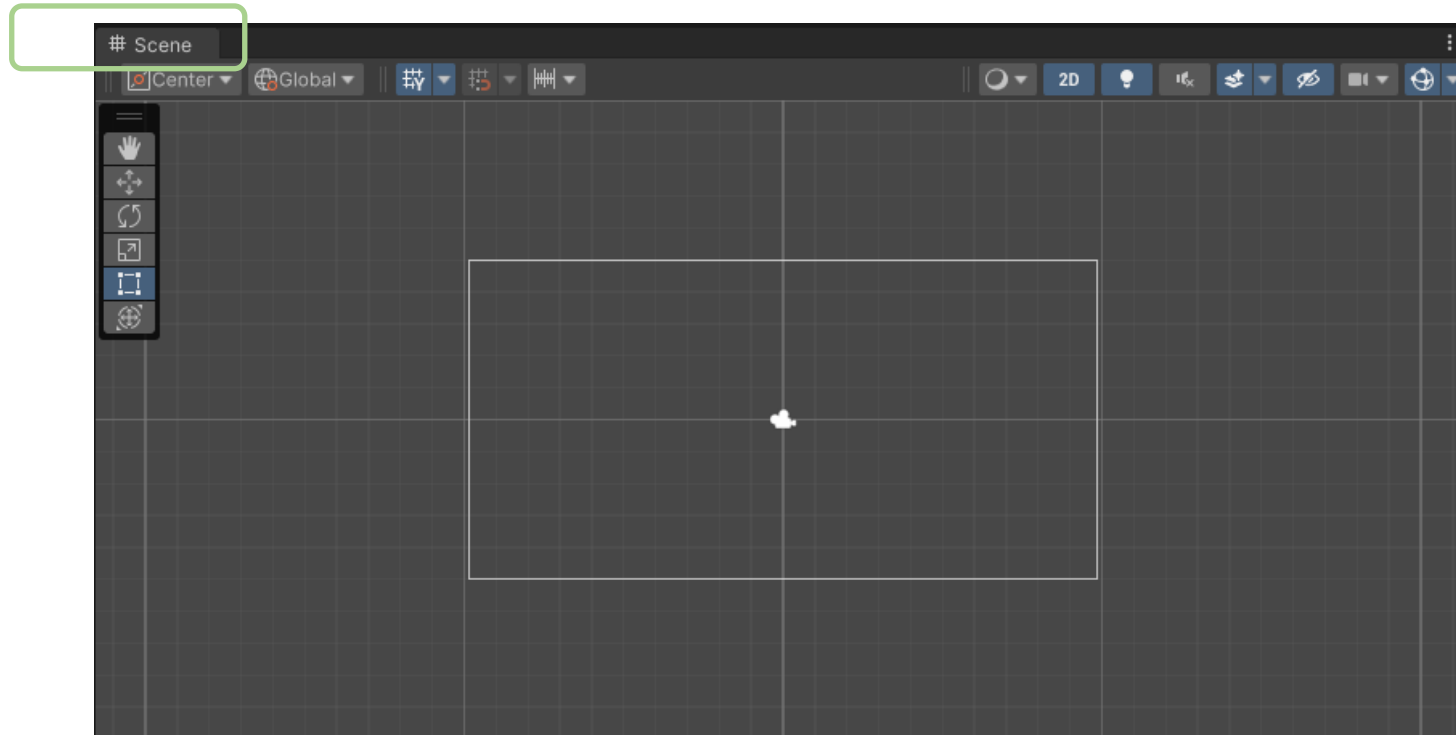


다양한 윈도 레이아웃 테스트



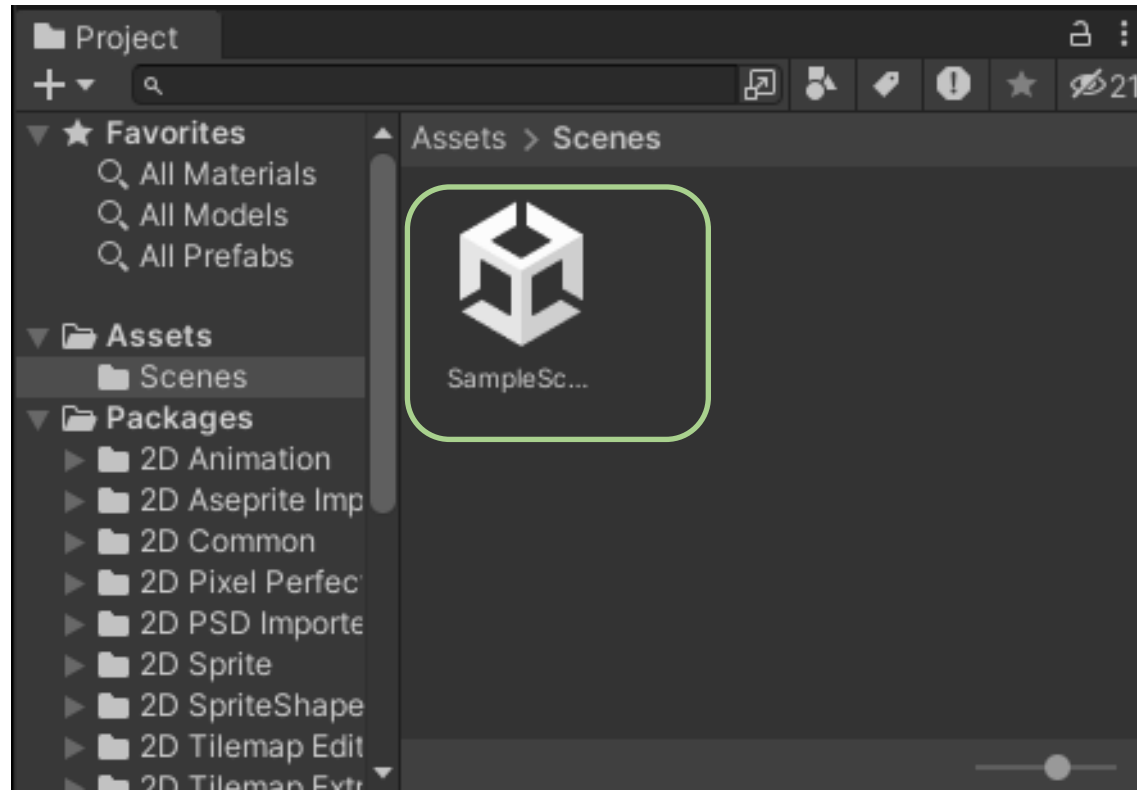
장면(Scene) – 게임 레벨

- 장면은 유니티에서 사용되는 일종의 Asset
- 장면에는 유니티 게임 프로젝트의 전부 혹은 일부 콘텐츠가 담김
- 유니티 프로젝트에서 개발은 이 장면 단위로 이루어짐

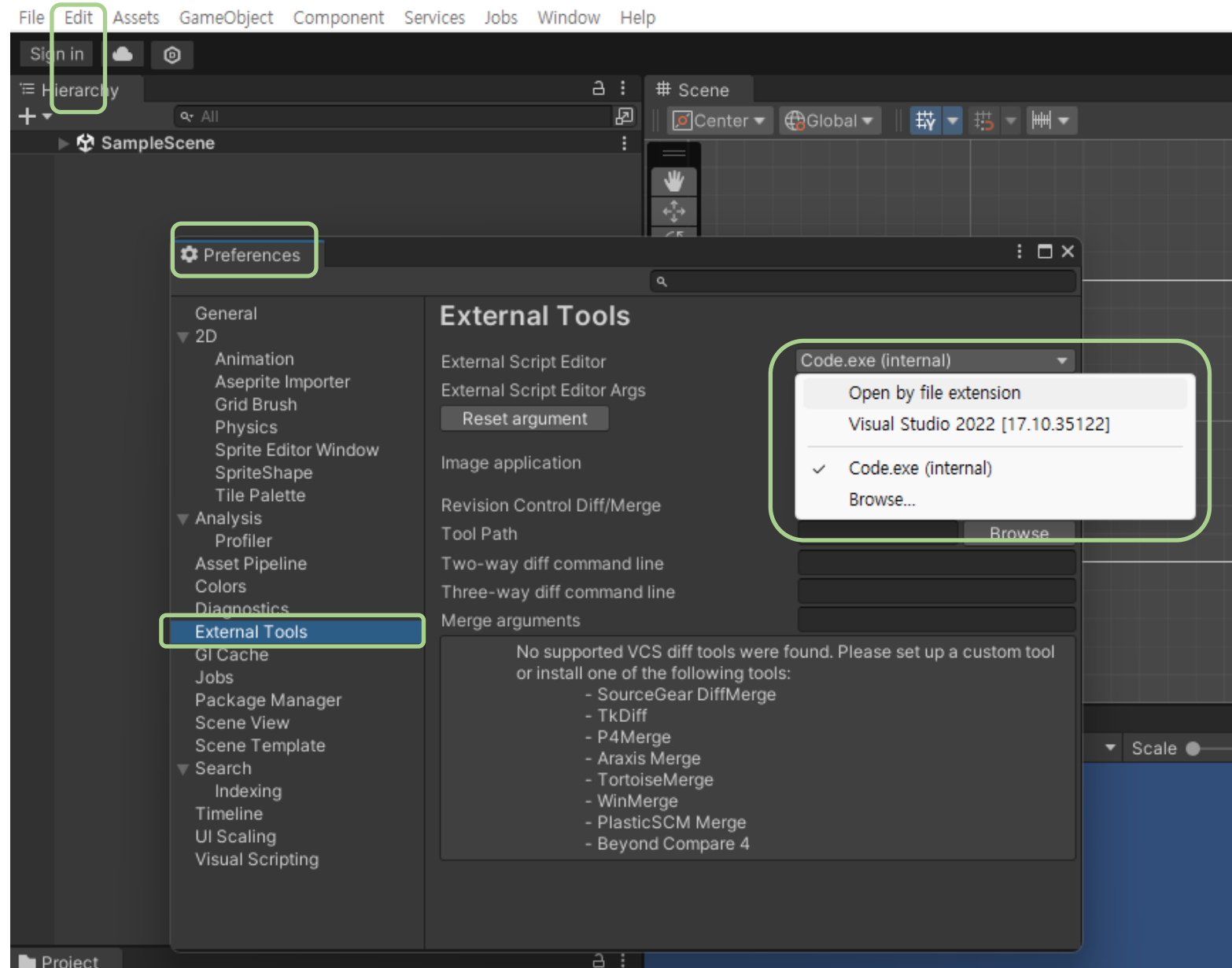


기본 장면 – SampleScene

- 유니티 프로젝트를 생성하면 SampleScene이 생성됨



설정

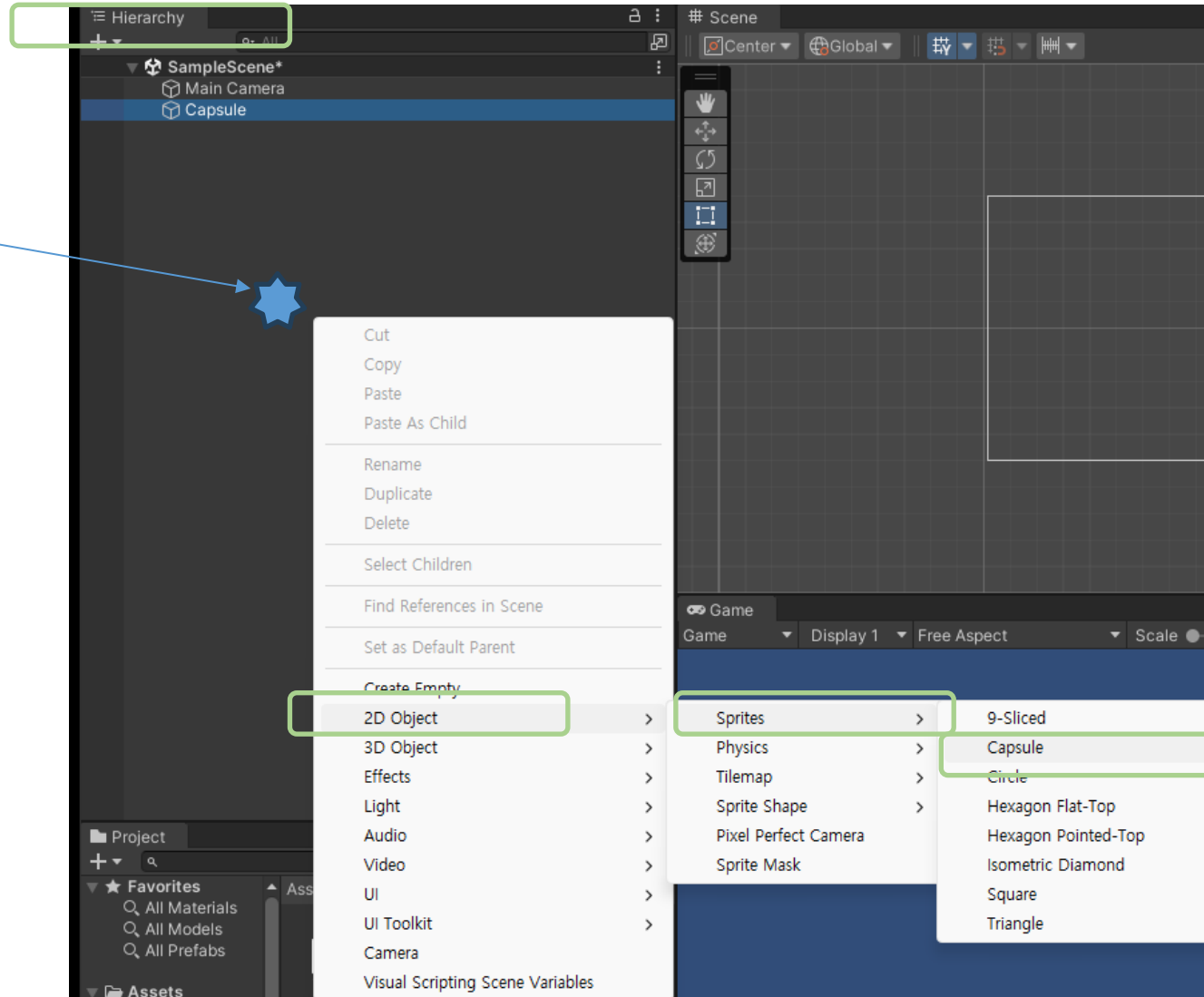


간단한 객체를 넣고 움직여 보자

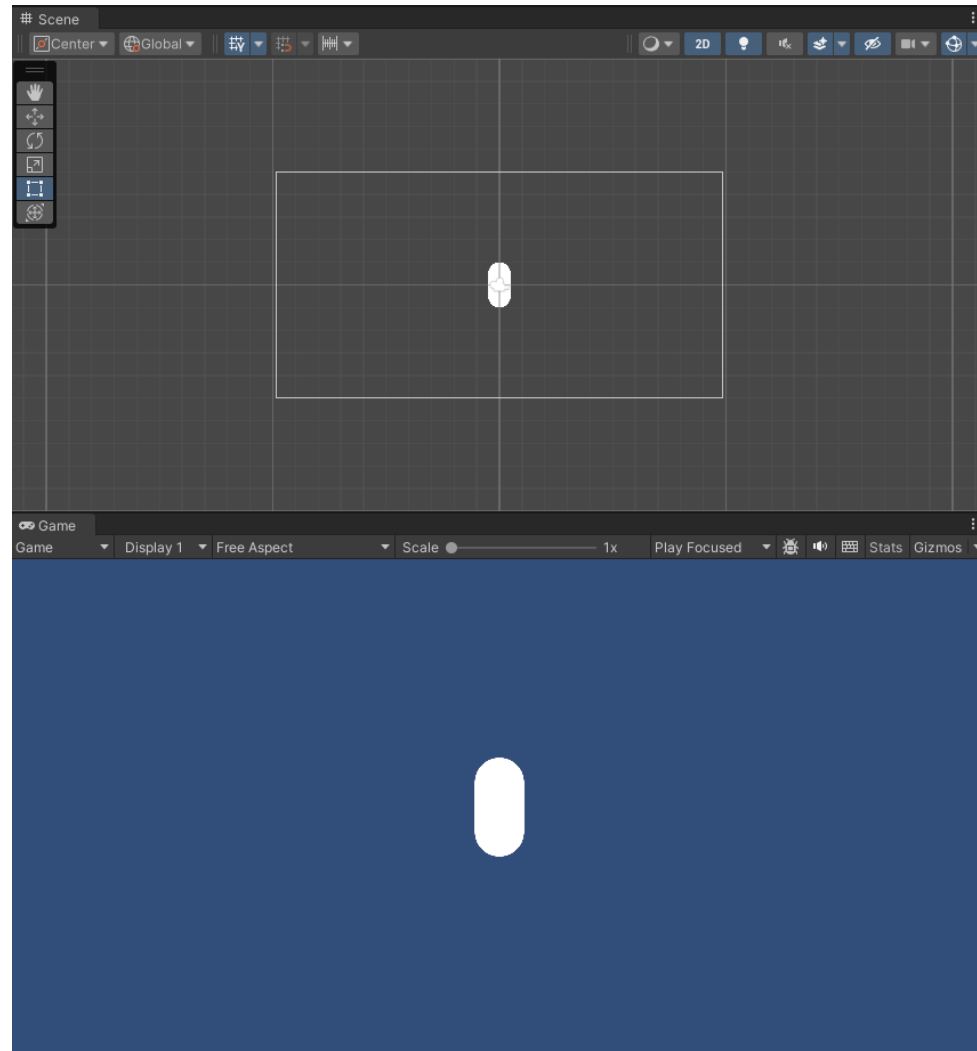
- 2D 게임 → 스프라이트 객체 사용
 - 캡슐 스프라이트 추가
 - 캡슐 스프라이트에 C# 코드 연결
 - 코딩을 통해 스프라이트 움직이기

캡슐 스프라이트 추가

Right click!



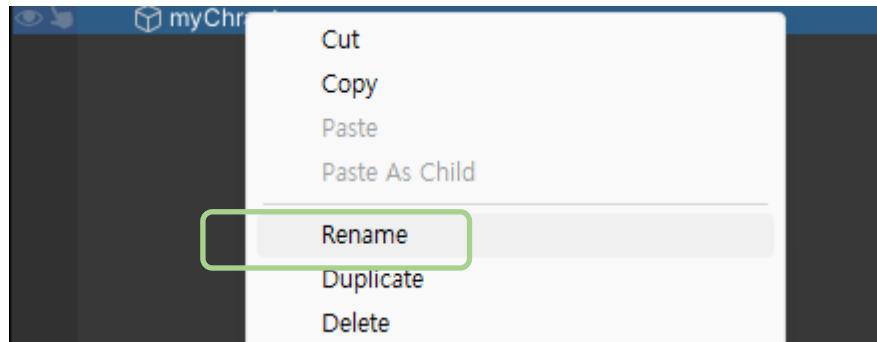
캡슐 스프라이트 확인



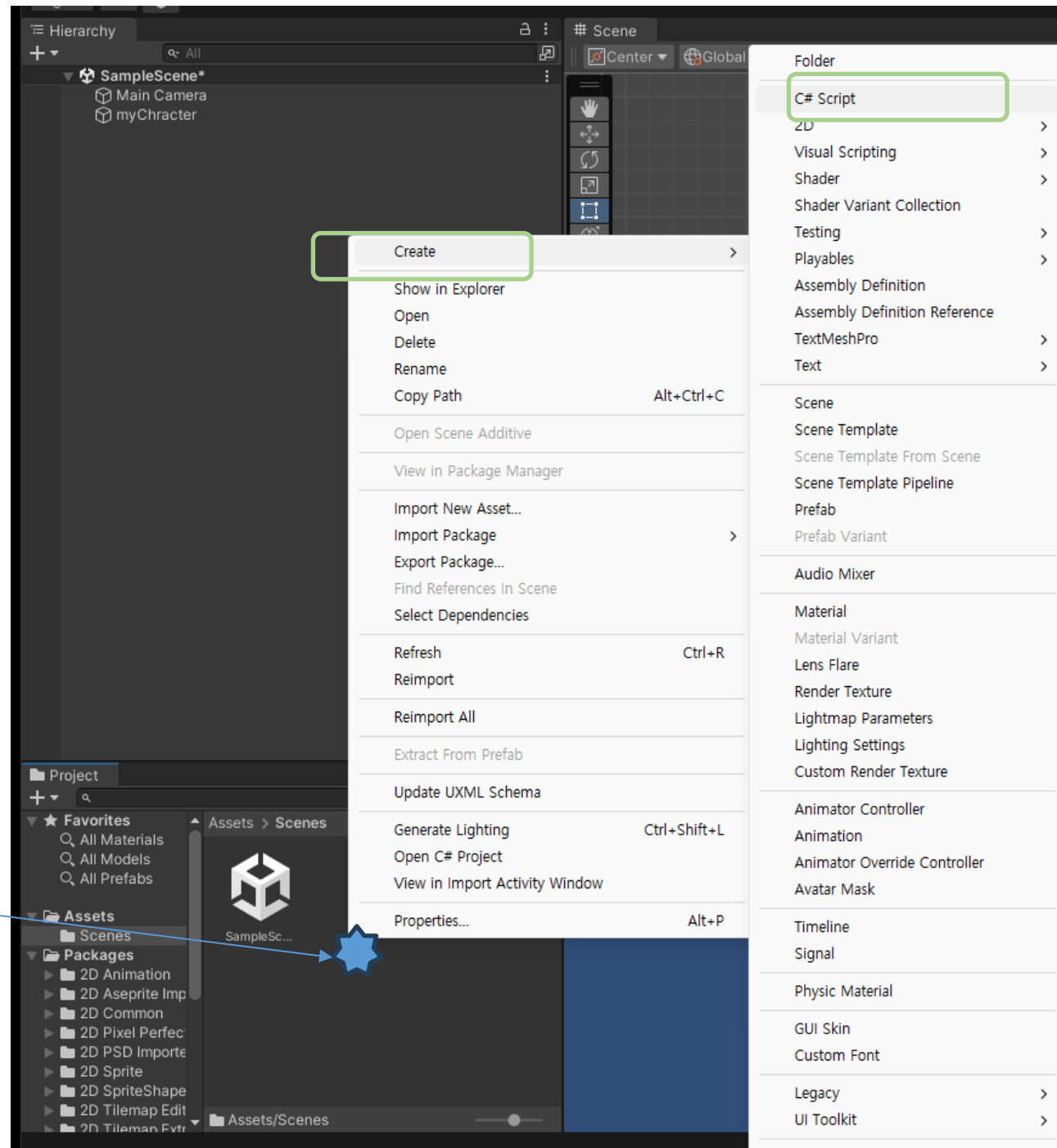
캡슐 스프라이트 움직이기 위한 “method” 사용

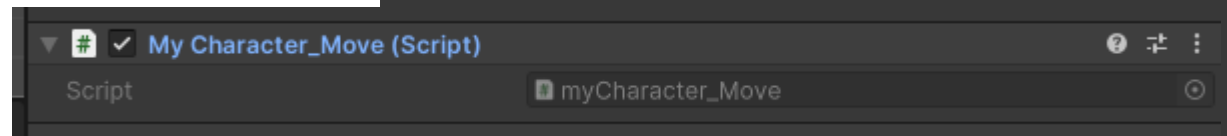
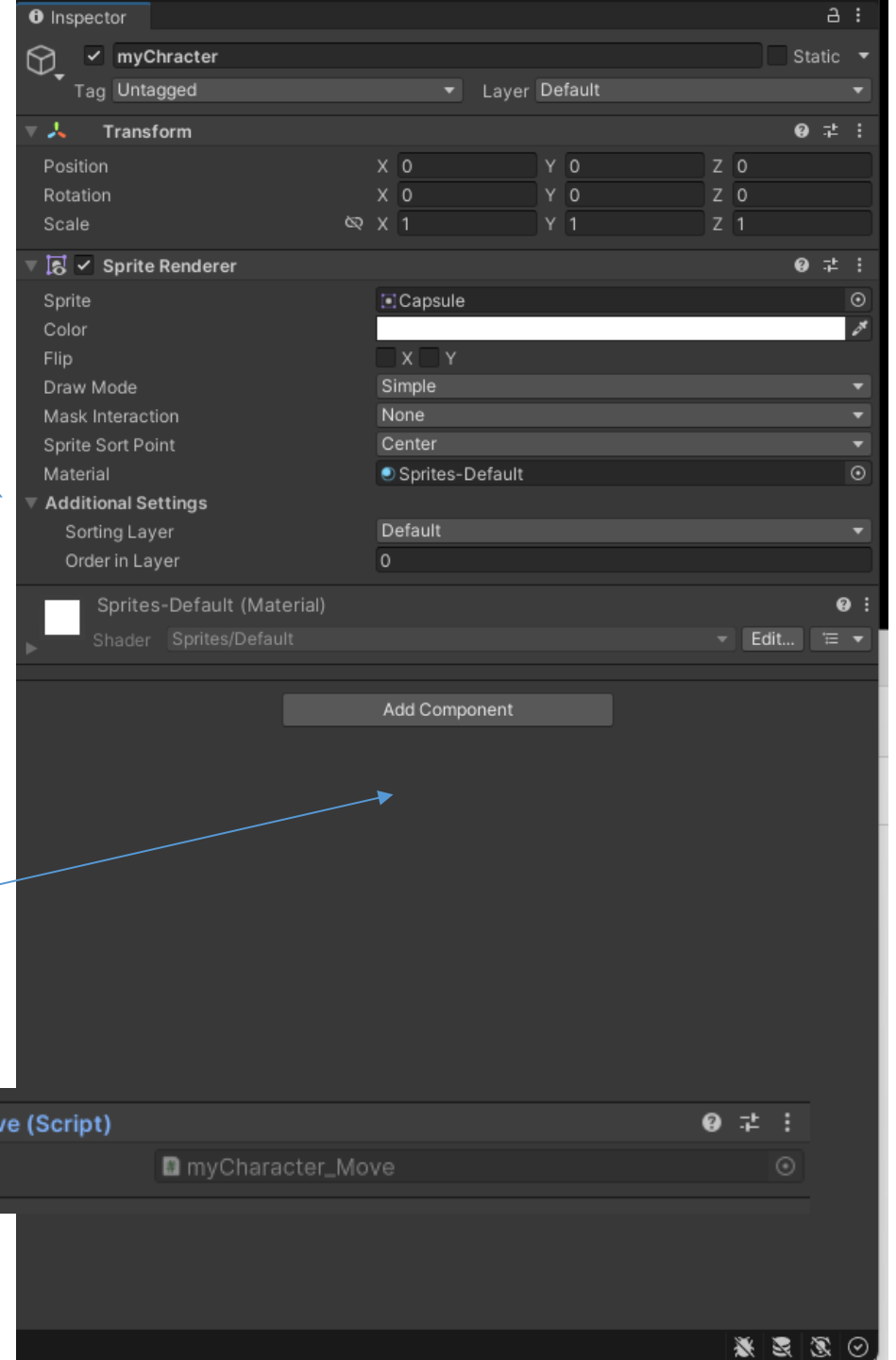
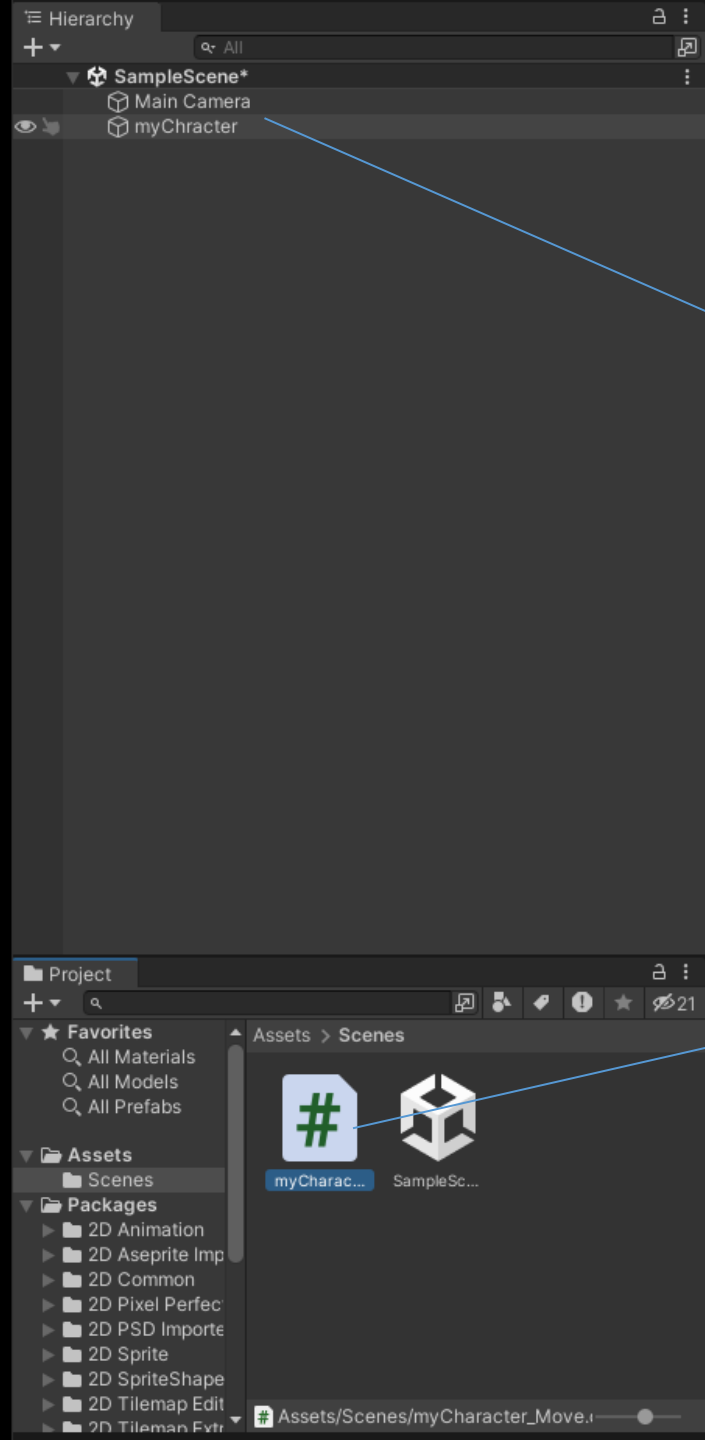
- 유니티의 메소드
 - 우리의 게임이 무엇인가를 할 수 있도록 하는 코드의 덩어리를 실행시키는 도구
 - 함수와 비슷한 개념 – 메소드는 객체에 딸린 함수
- 사용가능한 메소드
 - 유니티에서 제공하는 메소드
 - 우리가 직접 만든 메소드
- 메소드를 만들고 실행하는 것
 - 만들기: 무엇인가를 할 수 있는 코드를 메소드로 작성해 두는 것
 - 실행: 만들어 놓은 메소드를 호출하여 해당 동작이 이루어지게 하는 것

스크립트를 연결하기



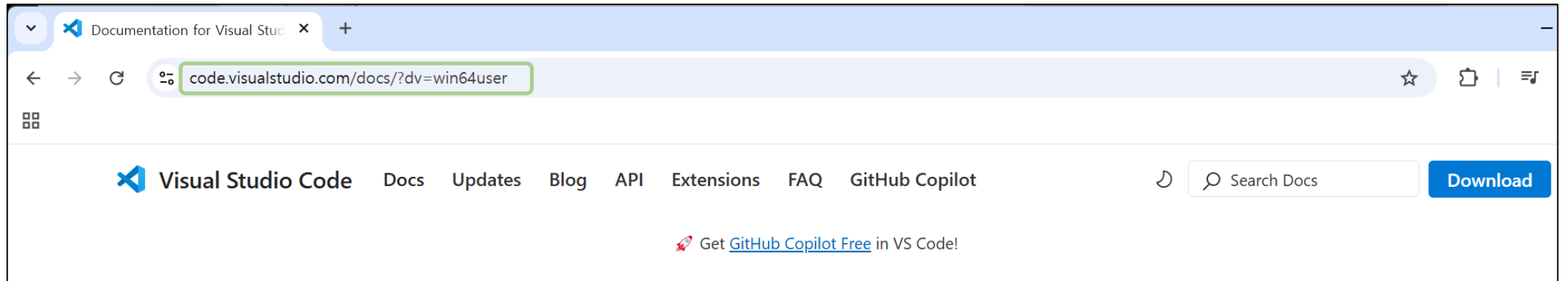
Right click!





스크립트를 수정해 보자

- 스크립트를 두 번 클릭
 - 연결된 외부 개발 도구가 열림
 - Visual Studio Code를 연결해 두었음
 - 열리지 않는 경우에는 도구가 설치되지 않은 상태임
- 필요한 경우 설치



필요한 Extension

The screenshot shows the Visual Studio Code interface with the C# Dev Kit extension installed. The left sidebar displays the 'EXTENSIONS: MARKETPLACE' view, where the 'C# Dev Kit' extension by Microsoft is highlighted with a green box. The main editor area shows the 'C# Dev Kit' extension details page, which includes the extension's icon, name, publisher (Microsoft), version (8,444,992), and a 4.5-star rating (157 reviews). The page also features buttons for 'Disable', 'Uninstall', and 'Switch to Pre-Release Version', along with an 'Auto Update' checkbox. The 'DETAILS' tab is selected, showing a description of the extension and a list of features.

EXTENSIONS: MARKETPLACE

- C# Dev Kit**
Official C# extension from Microsoft
Microsoft
- C#**
Base language support for C#
Microsoft
- IntelliCode for C# Dev Kit**
AI-assisted development for C# Dev Kit
Microsoft
- [DEPRECATED] C# Format Revolve**
[DEPRECATED] Fix format of usings / indents / braces
Przemysław Orłowski
- C# Extensions**
C# IDE Extensions for VSCode
jchannon
- C# Extensions**
C# IDE Extensions for VSCode
JosKreativ
- C# XML Documentation Comments**
Generate C# XML documentation comments for ///
Keisuke Kato
- C# Snippets**
C# Snippets for Visual Studio Code
Jorge Serrano
- Auto-Using for C#**
Provides intellisense for and imports references from al...
Fudge
- C# Namespace Autocompletion**
C# Namespace Autocompletion for Visual Studio Code
Jorge Serrano

C# Dev Kit
Microsoft [microsoft.com](#) | 8,444,992 | ★★★★★ (157)
Official C# extension from Microsoft

[Disable](#) [Uninstall](#) [Switch to Pre-Release Version](#) ☒ Auto Update

C# Dev Kit for Visual Studio Code

C# Dev Kit helps you manage your code with a solution explorer and test your code with integrated unit test discovery and execution, elevating your C# development experience wherever you like to develop (Windows, macOS, Linux, and even in a Codespace).

This extension builds on top of the great C# language capabilities provided by the [C# extension](#) and enhances your C# environment by adding a set of powerful tools and utilities that integrate natively with VS Code to help C# developers write, debug, and maintain their code faster and with fewer errors. Some of this new tooling includes but is not limited to:

- C# project and solution management via an integrated solution explorer
- Native testing environment to run and debug tests using the Test Explorer
- Roslyn-powered language service for best in-class C# language features such as code navigation, refactoring, semantic awareness, and more

Quick Start

- Install C# Dev Kit (The [C# extension](#) and the [.NET Install Tool](#) will automatically be installed).
- Open a folder/workspace that contains a C# project (.csproj) and the extension will activate, or use the ".NET: New Project..." command to create a new C# project.
- Check out the [C# Getting Started](#) documentation to learn more!

Feature List & Walkthrough

- Project System / Solution Explorer**
 - Solution Node Actions
 - Add Project
 - Build/Run Project

필요한 Extension

The screenshot displays the Visual Studio Code interface with the Unity extension marketplace open. The left sidebar shows a list of extensions, with 'Unity' highlighted. The main panel shows the details for the 'Unity' extension by Microsoft, which integrates Visual Studio Code with Unity. The extension is currently installed, and the 'Uninstall' button is visible. Below the extension details, there is a section titled 'Unity for Visual Studio Code' which describes the extension's purpose and lists its features.

Unity
Integrates Visual Studio Code with Unity
Microsoft

Debugger for Unity
Unity debugger extension
Unity Technologies

Unity Tools
Various tools to help with Unity development
Avin Zarlez

Unity Code Snippets
All snippets you need for Unity3D development
Kleber Silva

Unity Snippets
Unity MonoBehaviour function snippets with document...
Ycleptic Studios

eppzi! (C# theme for Unity)
Carefully designed colors with meanings.
eppzi!

Unity Toolbox
Awesome tools for VSCode & Unity
maxsroka

Unity Snippets Modified
Unity MonoBehaviour function snippets with document...
lionize

Unity Code Snippets
All snippets you need for Unity3D development
cemuka

Unity Dev Pack
Some extensions that will help you use VSCode with U...
fabriciohod

Unity Code
All snippets you need for Unity3D, Unity2D, UnityGame...
Bakken Bui-Gemini

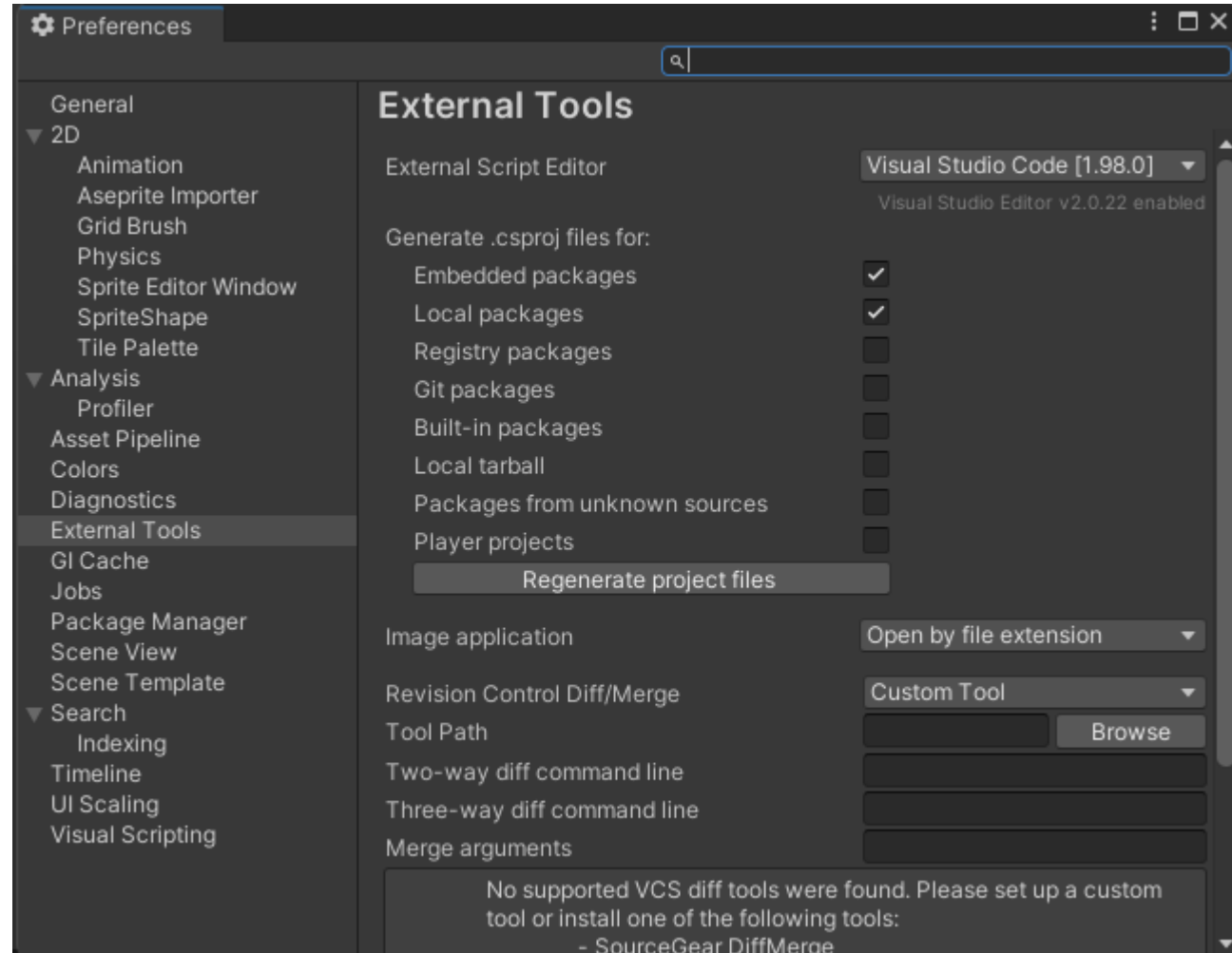
Unity for Visual Studio Code

The Unity extension provides C# developers with a lightweight and streamlined [Unity](#) development experience on Visual Studio Code. It builds on top of the rich C# capabilities provided by the [C# Dev Kit](#) and [C#](#) extensions. This extension integrates natively with Visual Studio Code and includes multiple productivity features including:

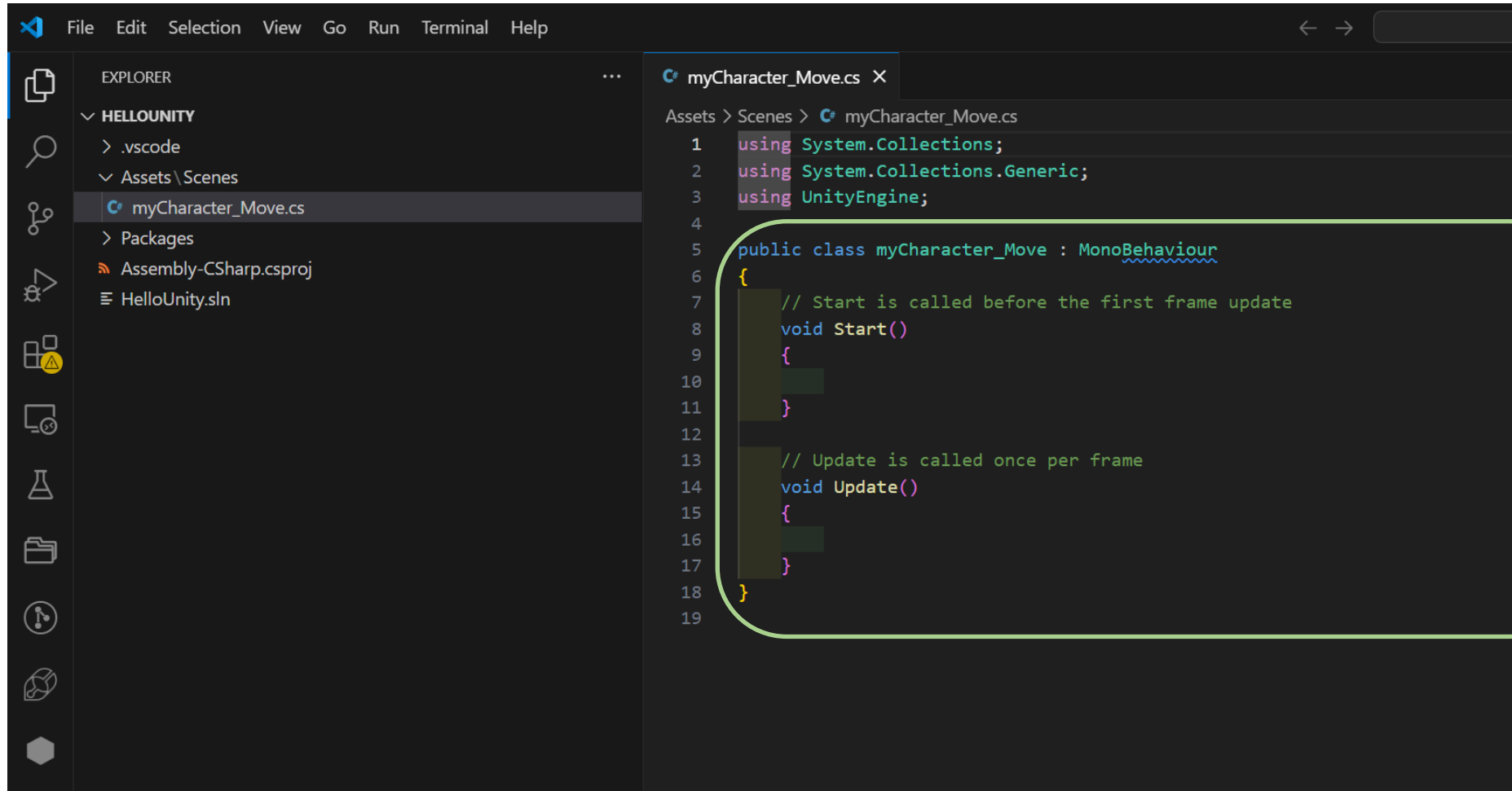
- A Unity debugger to debug your Unity editor and your Unity players.
- Unity specific C# analyzers and refactorings.
- Code coloration for Unity file formats (.asmdef, .shader, .uss, .uxml).

The screenshot also shows a preview of the Unity editor running a game, and a code editor displaying C# code for a Unity script.

Unity Preference



코드 편집

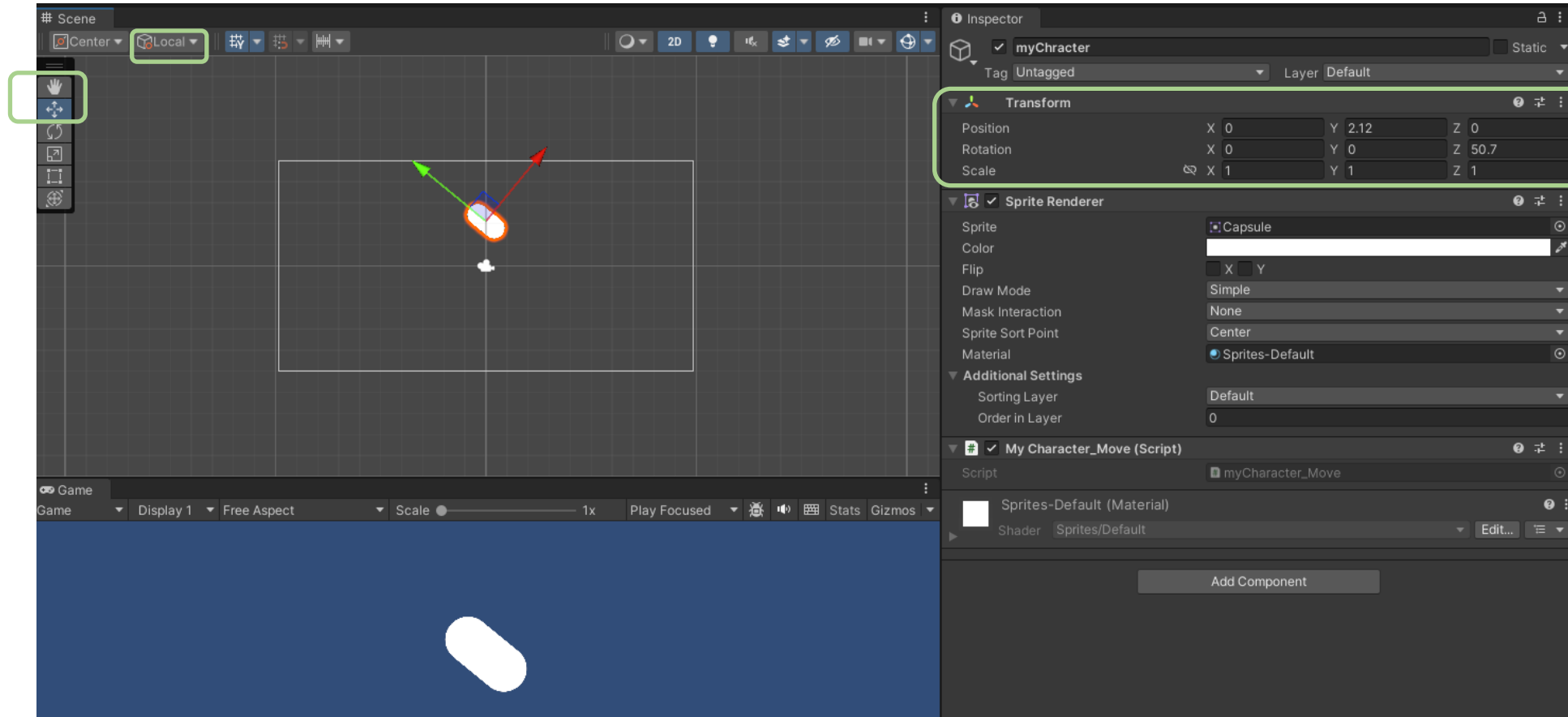


이 클래스가
우리의 캐릭터에
연결되어 있음

두 개의 클래스
Start, Update는
자동으로 호출되는
메소드

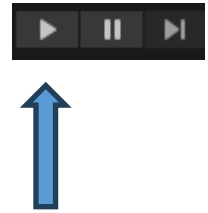
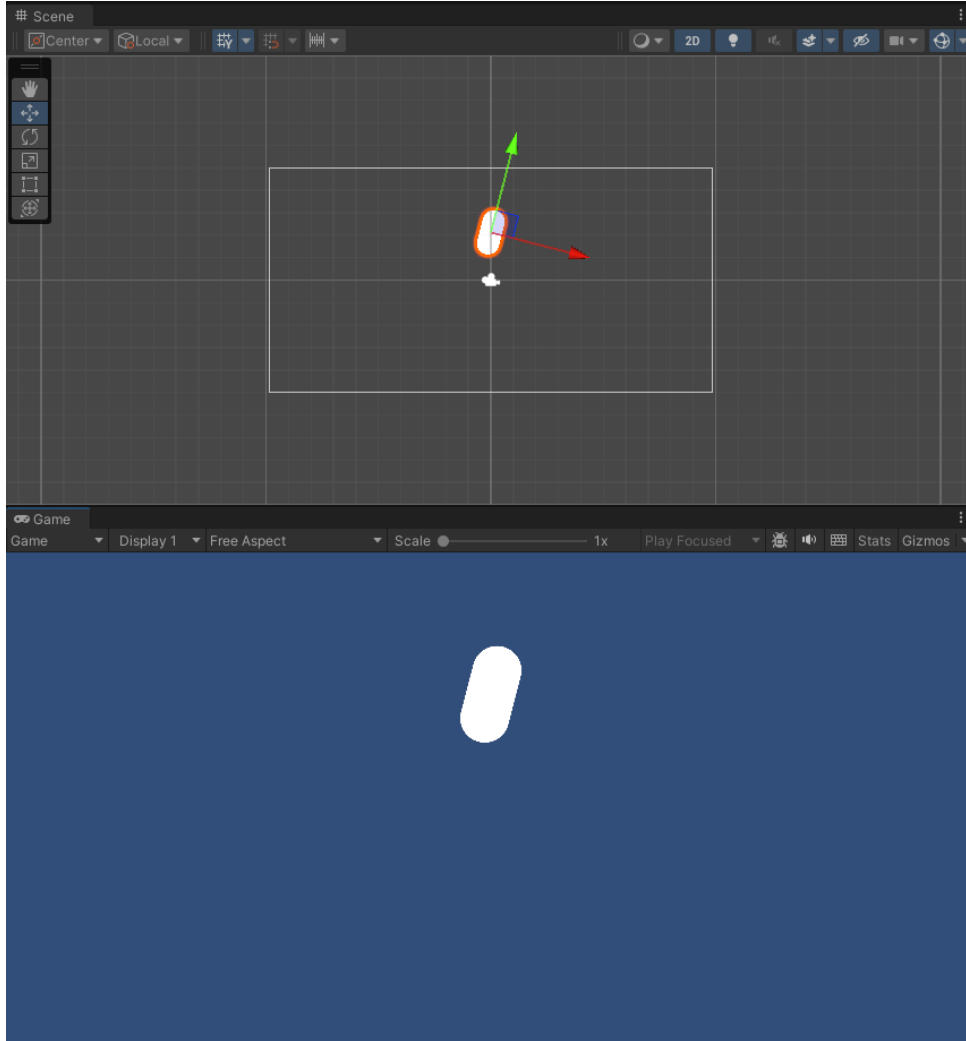
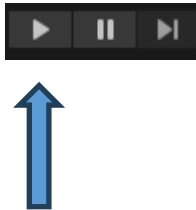
Start: 플레이될 때
Update: 매 프레임

인스펙터(Inspector)로 값의 변화 확인



움직이는 첫 프로젝트

Play 버튼을 누른다



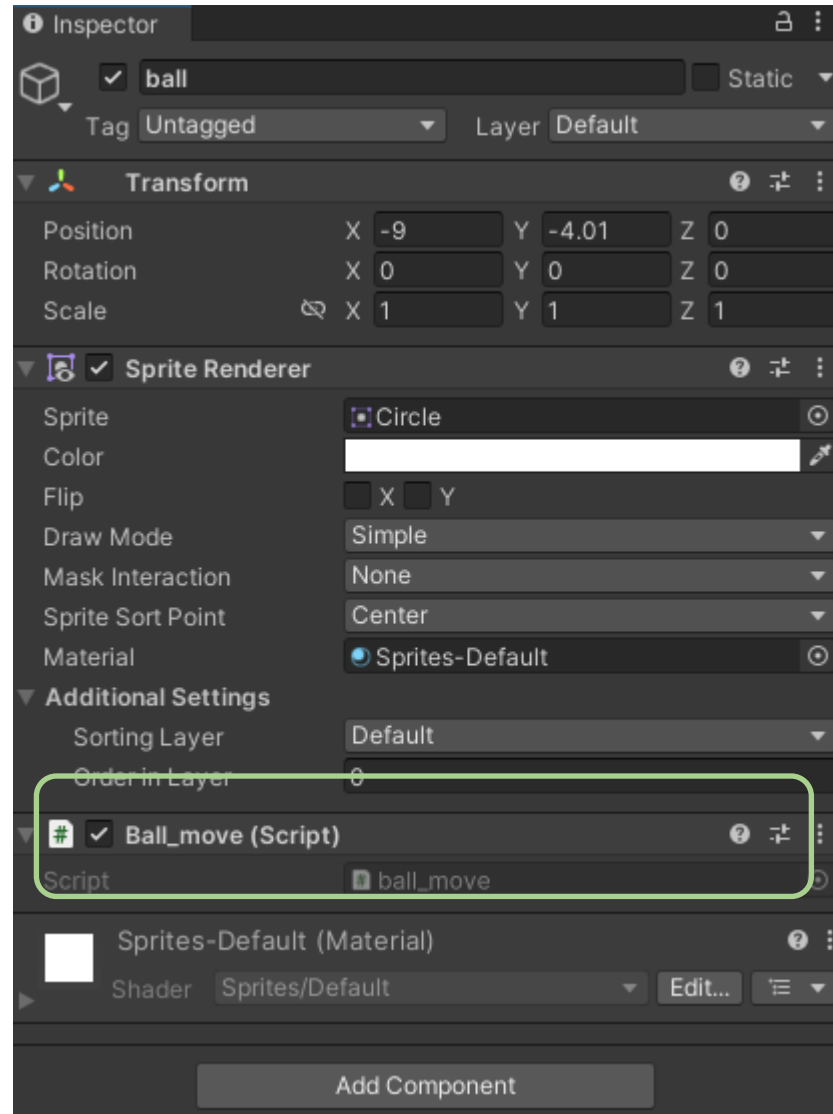
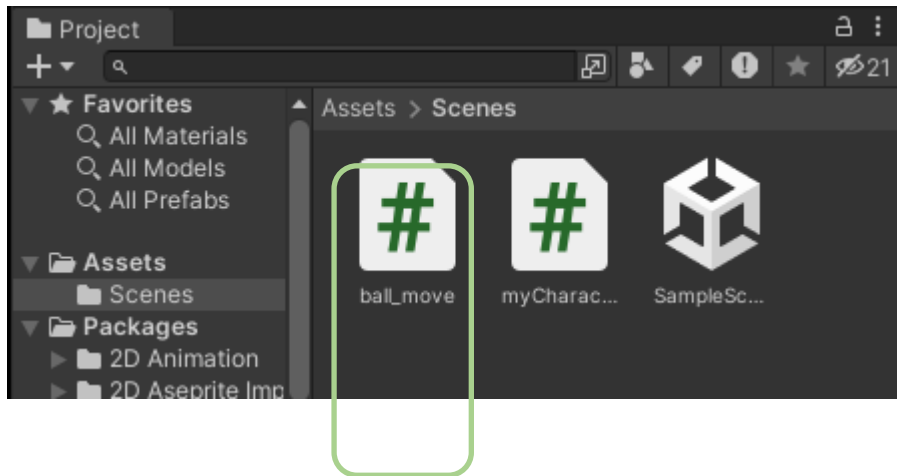
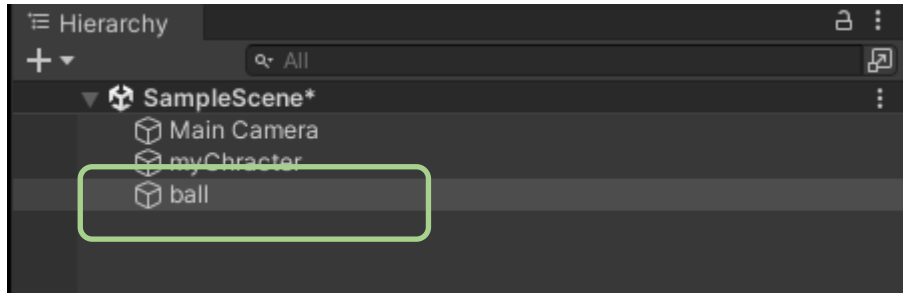
멈춰야지 변경이 가능

새로운 메소드: transform.Translate

```
13 // Update is called once per frame
    0 references
14 void Update()
15 {
16     transform.Rotate(0, 0.0f, 0.5f);
17     transform.Translate(0.0f, 0.01f, 0.0f);
18 }
19 }
20
```



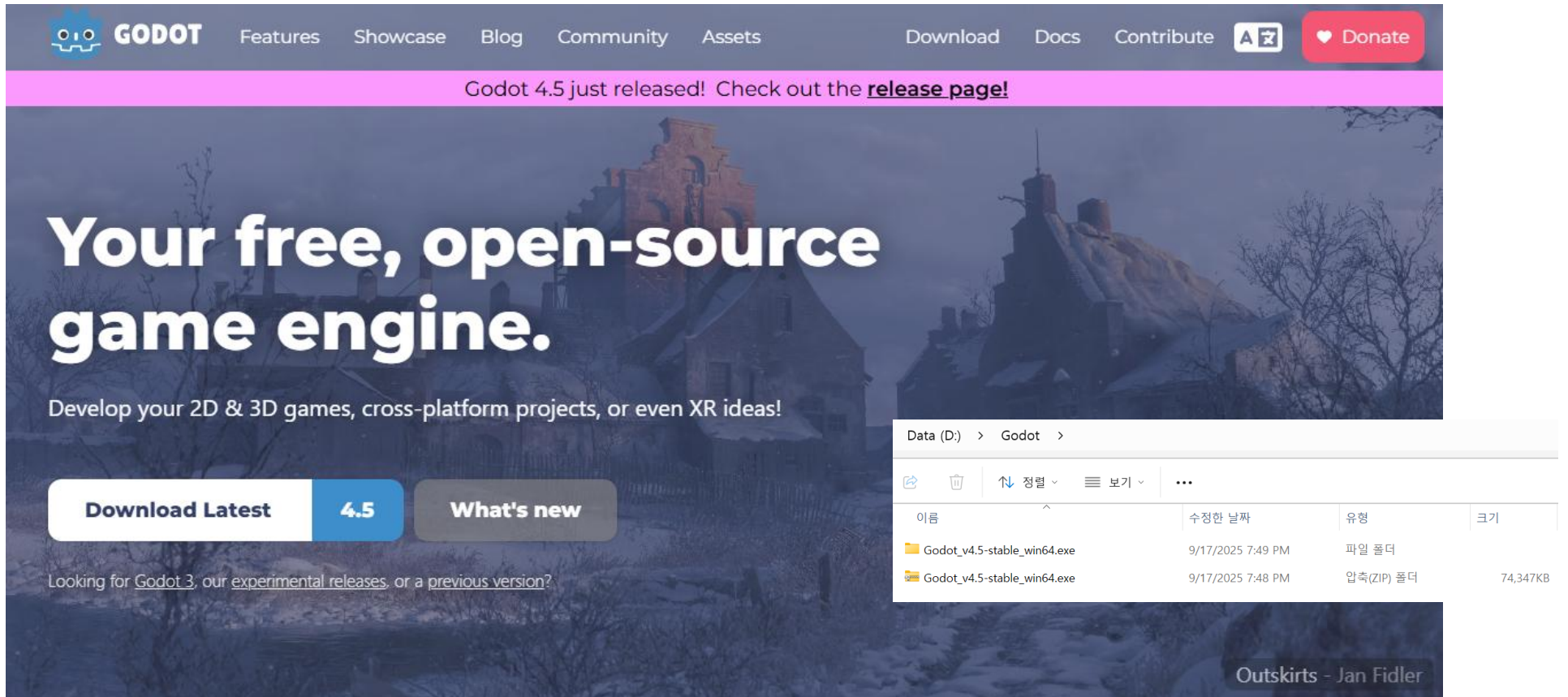
새로운 캐릭터 만들어 보기





엔진 다운로드 사이트

<https://godotengine.org/>



The screenshot shows the Godot Engine website with a navigation bar containing links for Features, Showcase, Blog, Community, Assets, Download, Docs, Contribute, and a Donate button. A pink banner announces "Godot 4.5 just released! Check out the [release page!](#)". The main heading reads "Your free, open-source game engine." followed by the tagline "Develop your 2D & 3D games, cross-platform projects, or even XR ideas!". Below this are buttons for "Download Latest", "4.5", and "What's new". A link at the bottom suggests looking for Godot 3, experimental releases, or a previous version. The background features a 3D game scene titled "Outskirts - Jan Fidler".

Overlaid on the bottom right is a Windows File Explorer window showing the contents of the "Data (D:) > Godot" directory. It contains two entries:

이름	수정된 날짜	유형	크기
Godot_v4.5-stable_win64.exe	9/17/2025 7:49 PM	파일 폴더	
Godot_v4.5-stable_win64.exe	9/17/2025 7:48 PM	압축(ZIP) 폴더	74,347KB

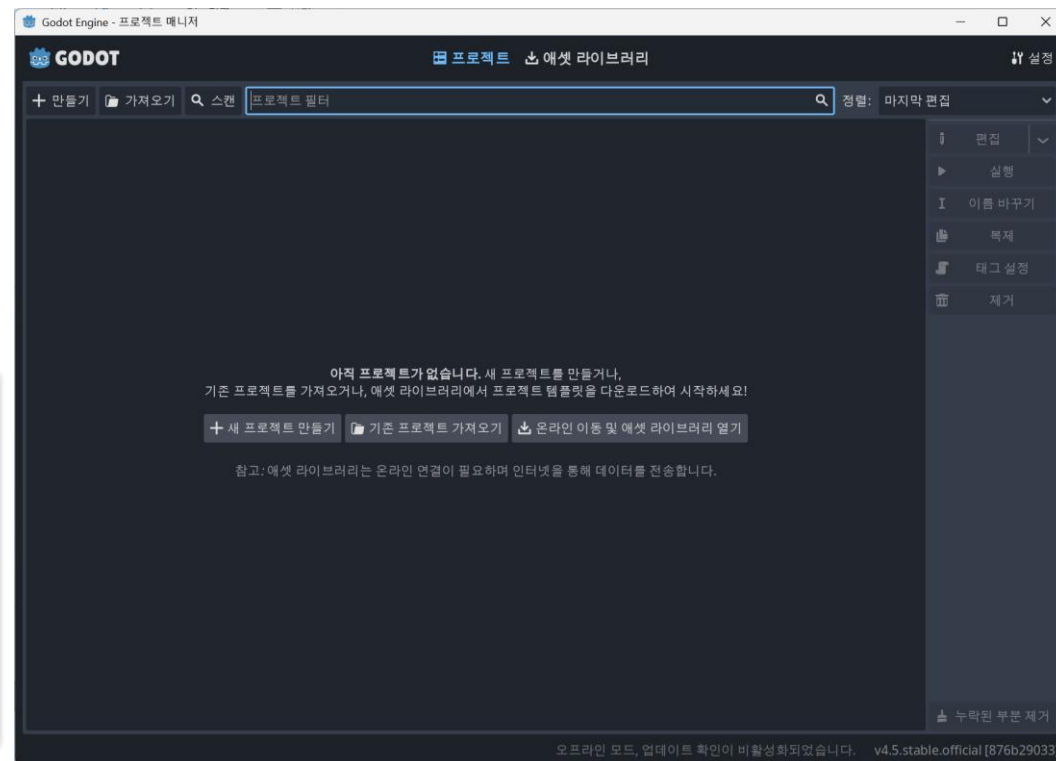
초간단 설치와 실행

다운로드된 압축파일을 풀기만 하면 됨

- 실행 파일 실행하면 엔진 실행됨

Data (D:) > Godot >			
이름 수정한 날짜 유형 크기			
Godot_v4.5-stable_win64.exe	9/17/2025 7:49 PM	파일 폴더	
Godot_v4.5-stable_win64.exe	9/17/2025 7:48 PM	압축(ZIP) 폴더	74,347KB

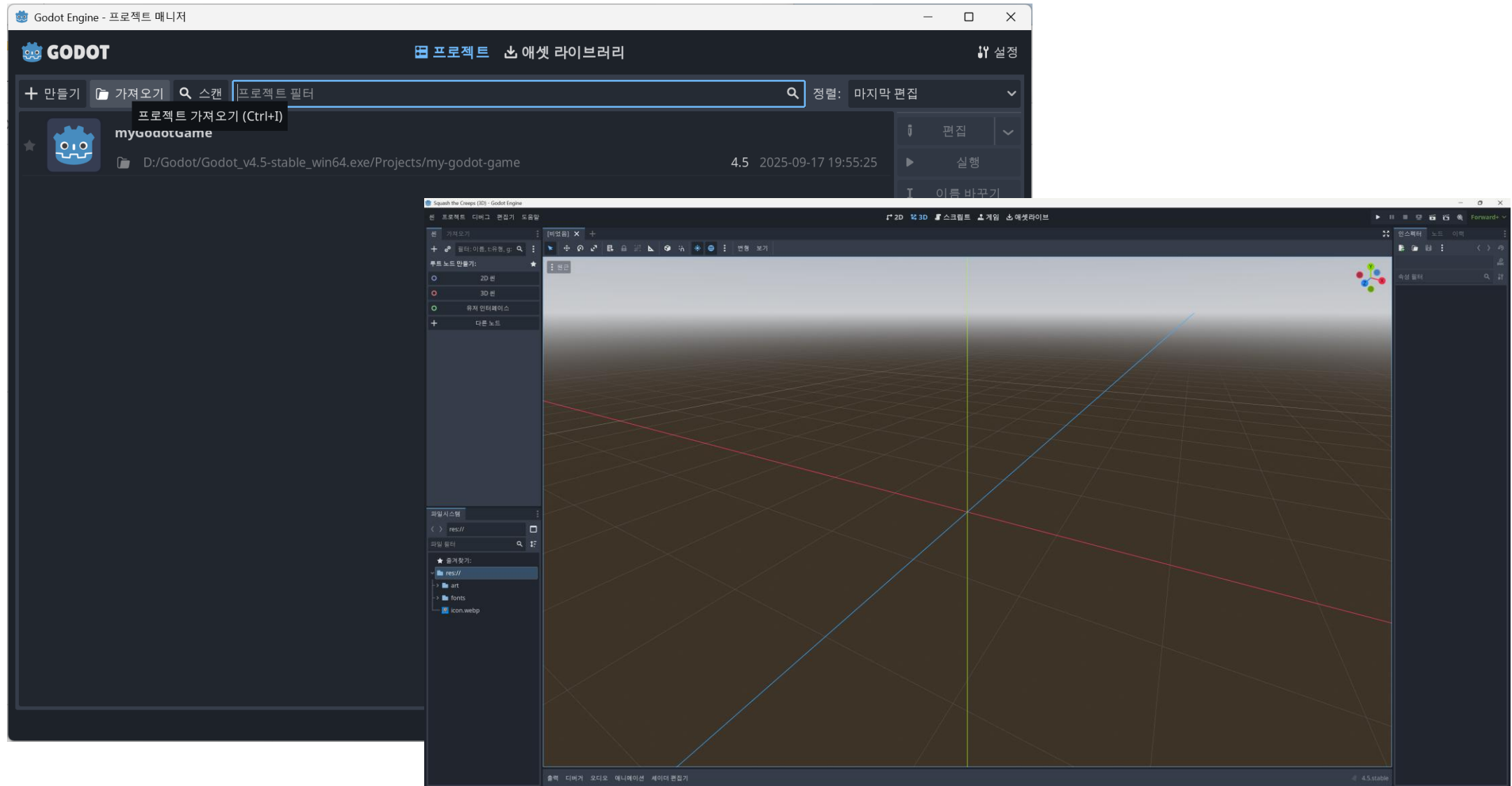
Data (D:) > Godot > Godot_v4.5-stable_win64.exe			
이름 수정한 날짜 유형 크기			
Godot_v4.5-stable_win64	9/17/2025 7:49 PM	응용 프로그램	159,010KB
Godot_v4.5-stable_win64_console	9/17/2025 7:49 PM	응용 프로그램	194KB



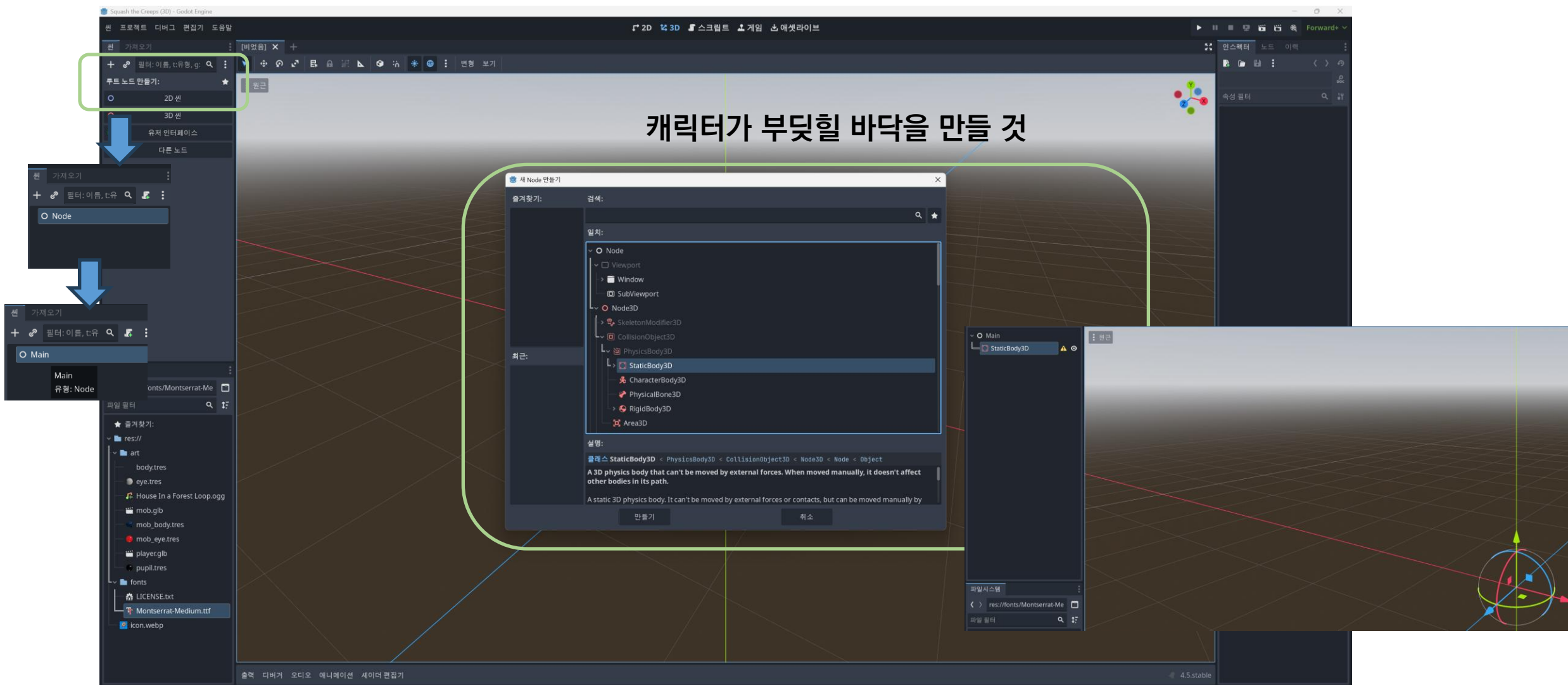
프로젝트 생성



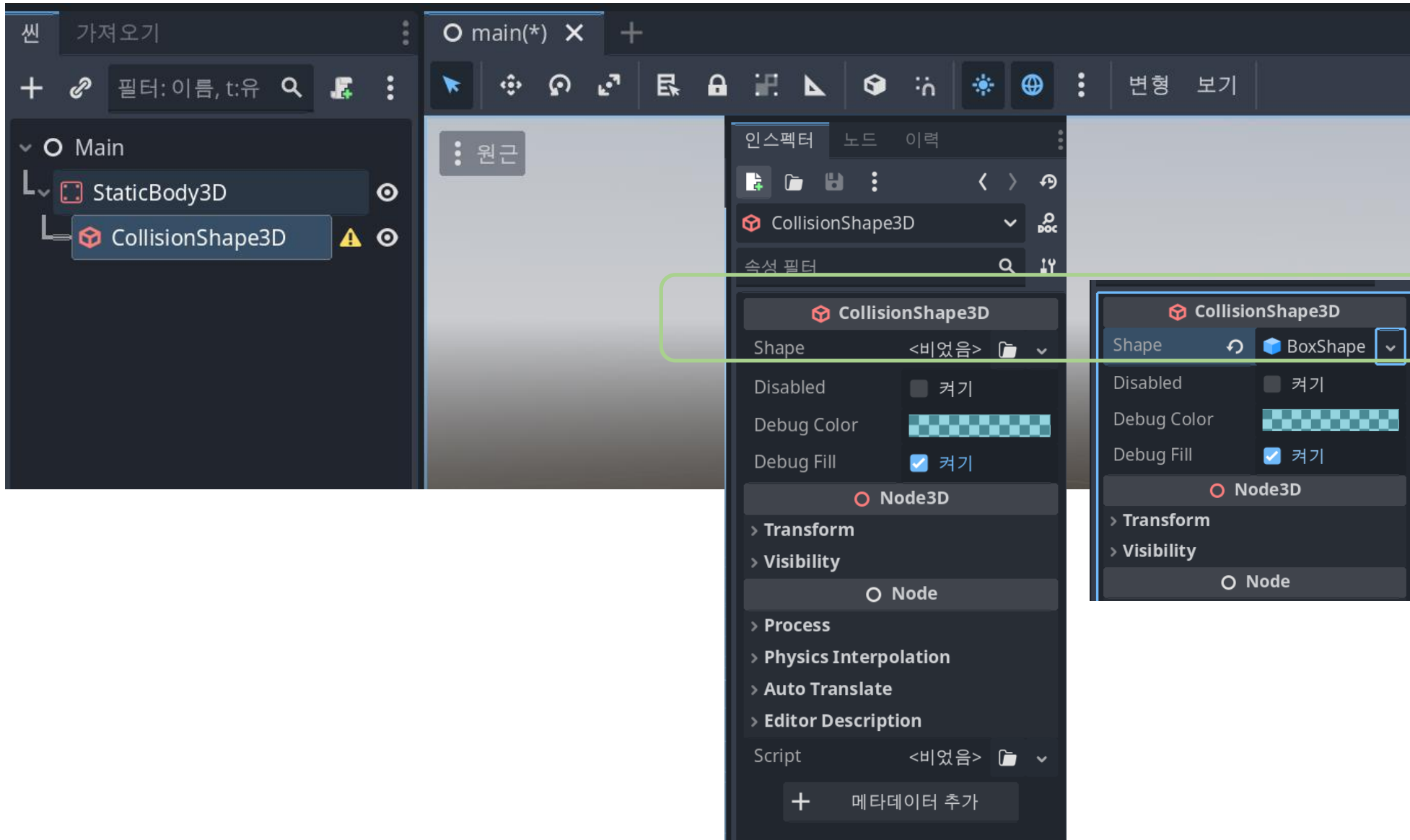
프로젝트 가져오기



메인 씬 생성 (루트 노드 생성) – 자식 노드 생성



충돌 처리

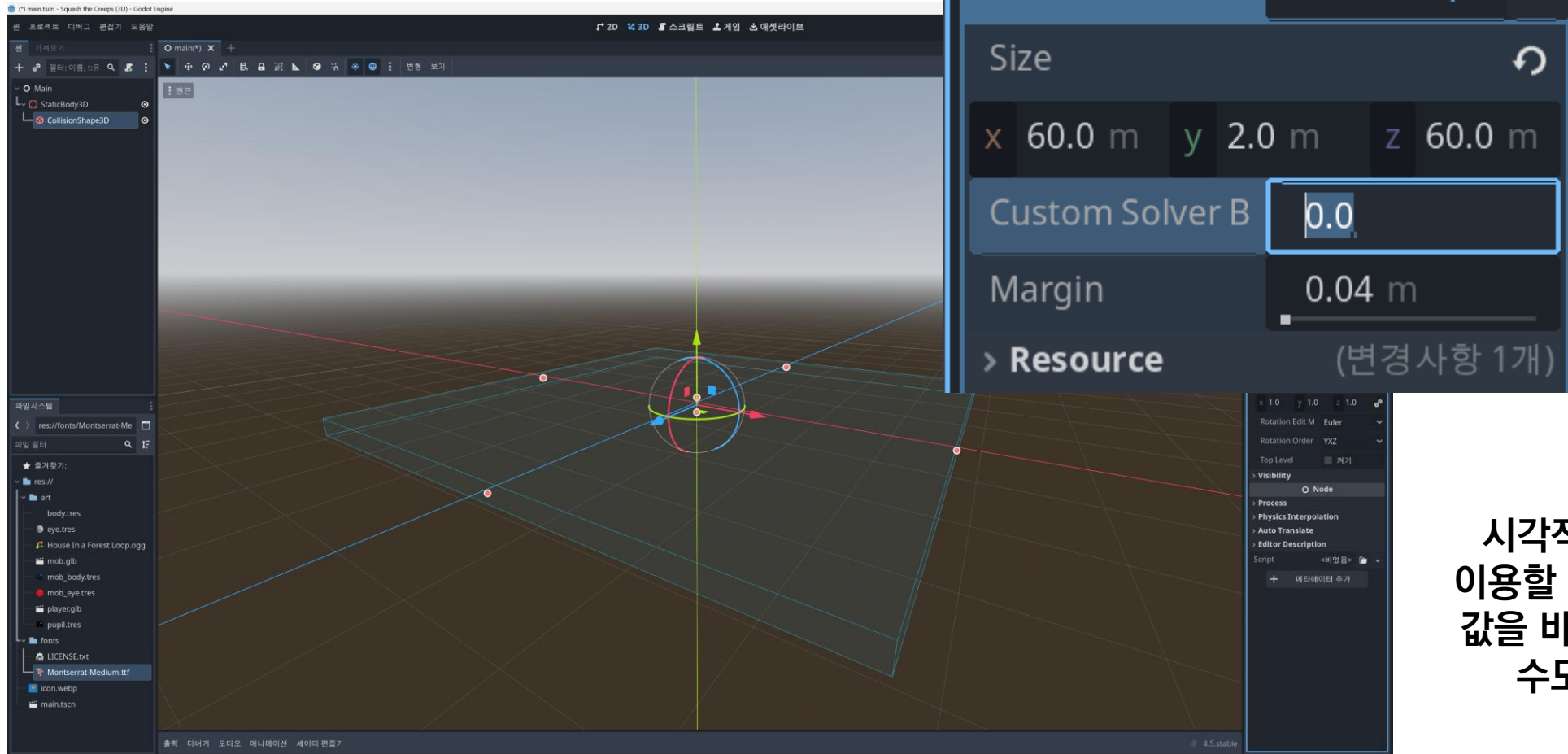


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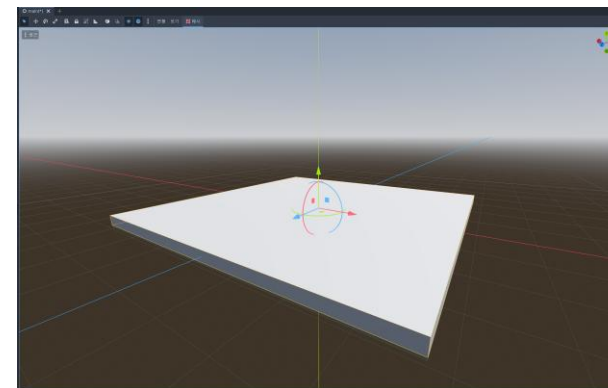
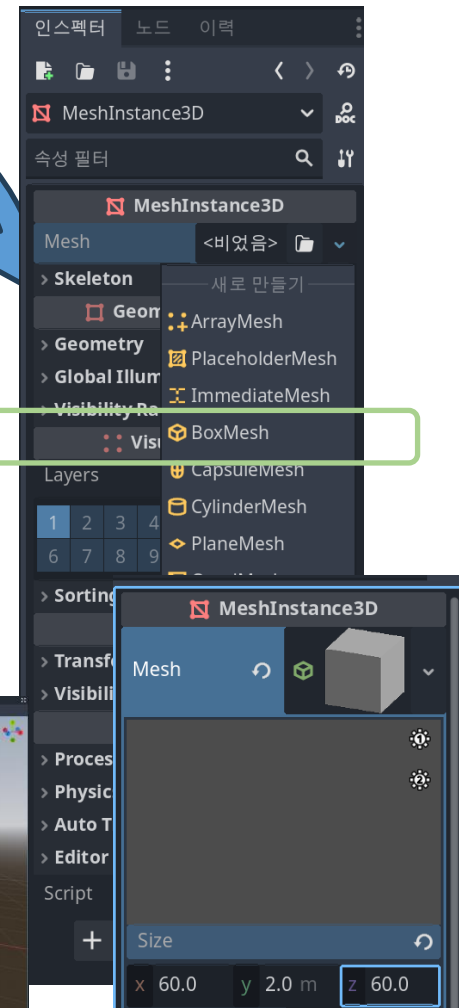
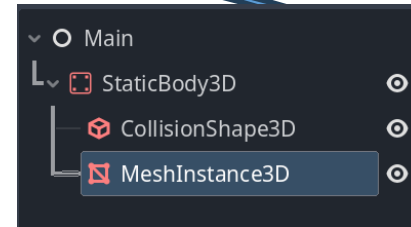
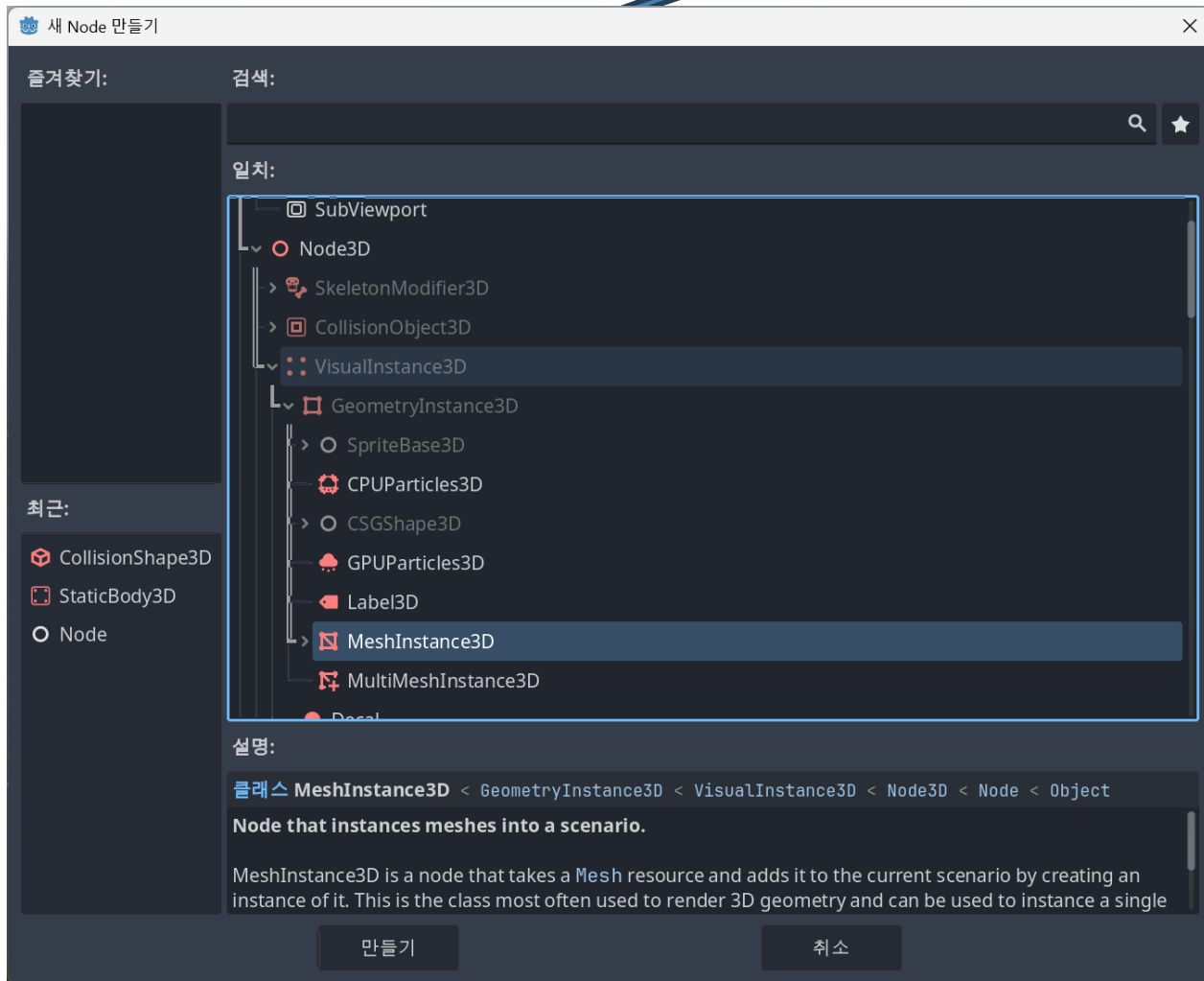
Start: 플레이될 때
Update: 매 프레임

충돌 박스 크기 조정

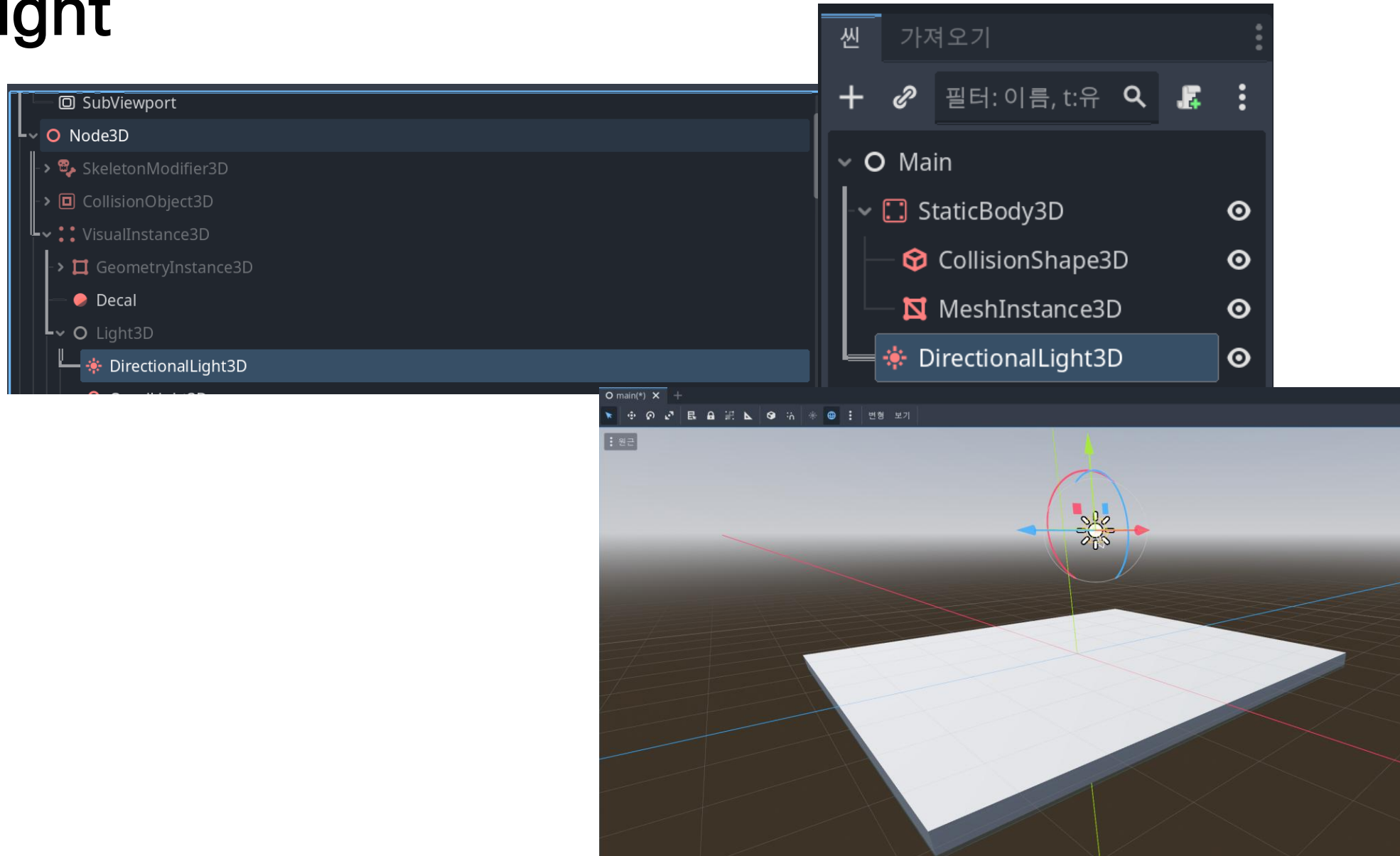


시각적 도구를
이용할 수도 있고,
값을 바로 지정할
수도 있음

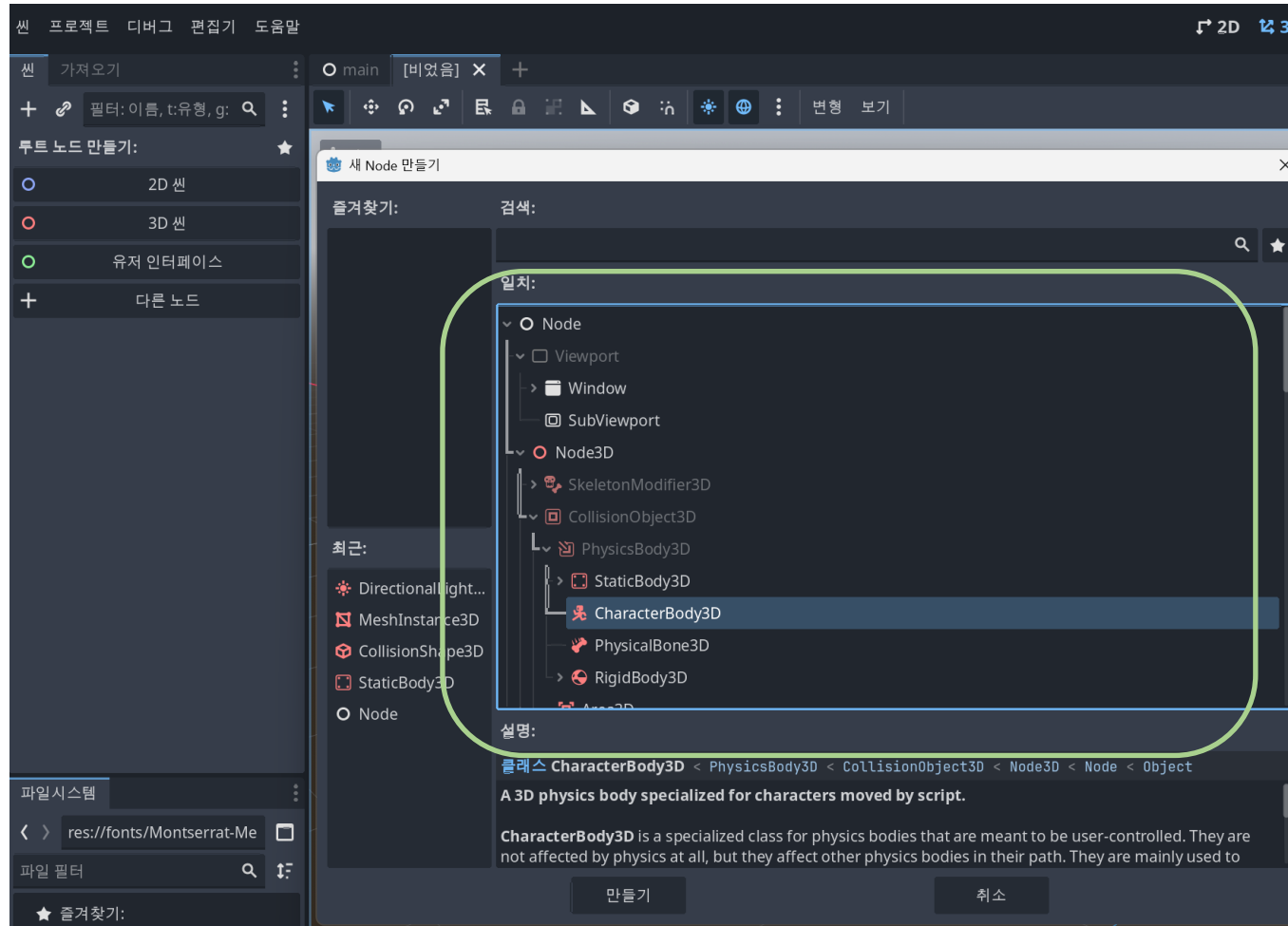
가시화 객체 추가



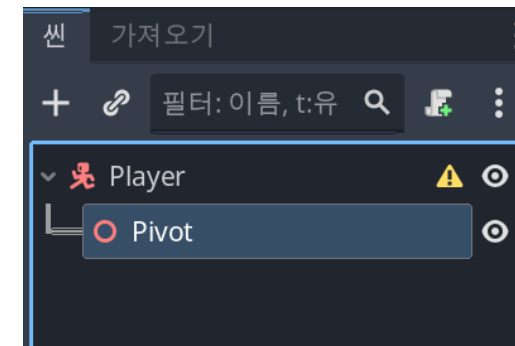
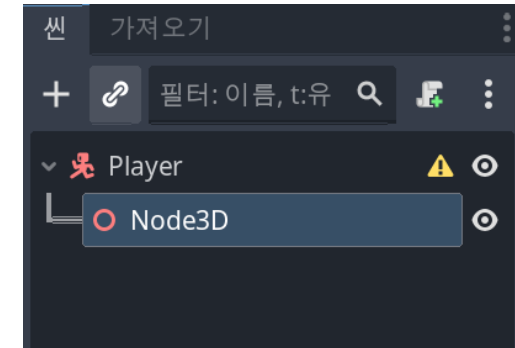
Light



플레이어 씬 새로 만들기

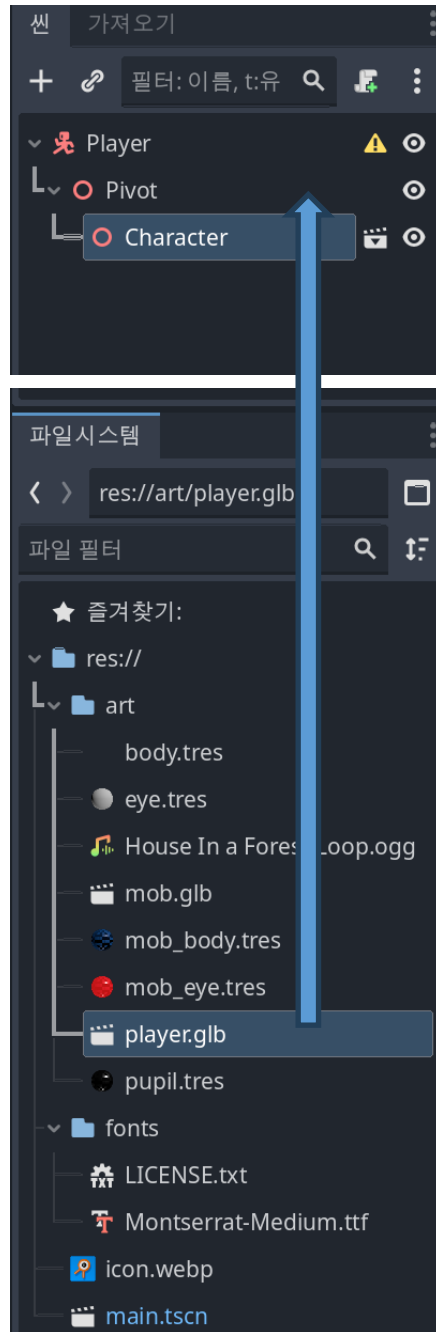


1. 씬 → 새로 만들기
2. 캐릭터 바디 추가
3. 노드 3D 자식 노드 추가
4. 노드 3D를 Pivot으로 이름 변경



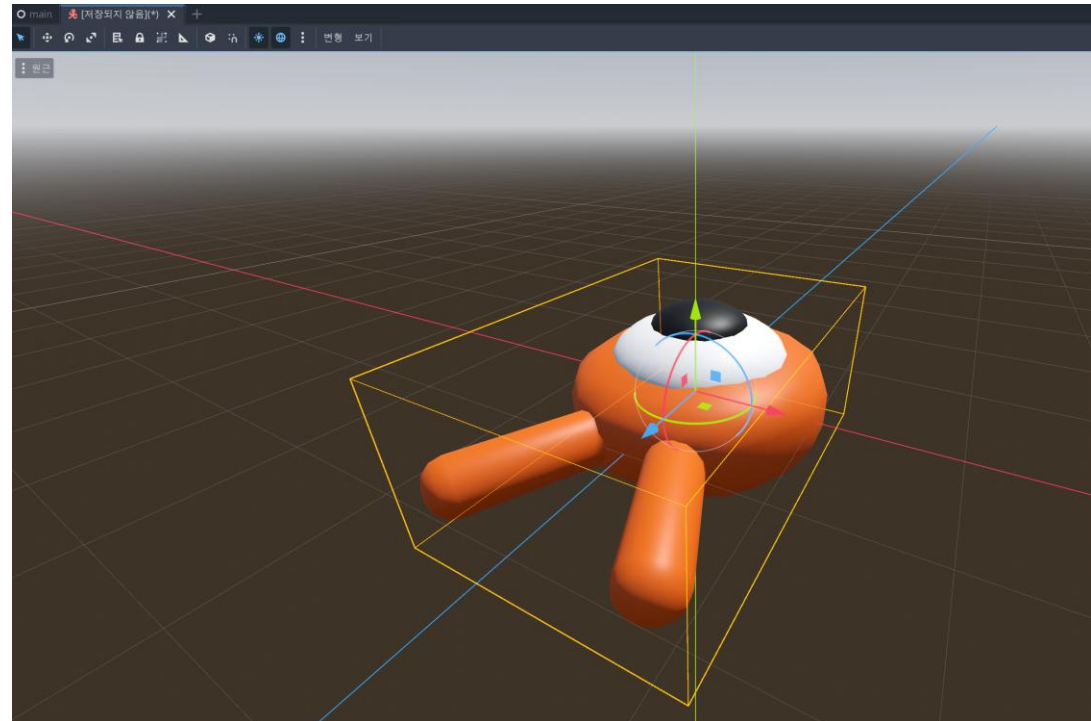
코드 편집

A **GLB file** is the **binary version of the GLTF (GL Transmission Format)**, which is an open standard file format for 3D models. It is designed to efficiently transmit and load 3D assets in real-time applications such as web, mobile, AR/VR, and games.

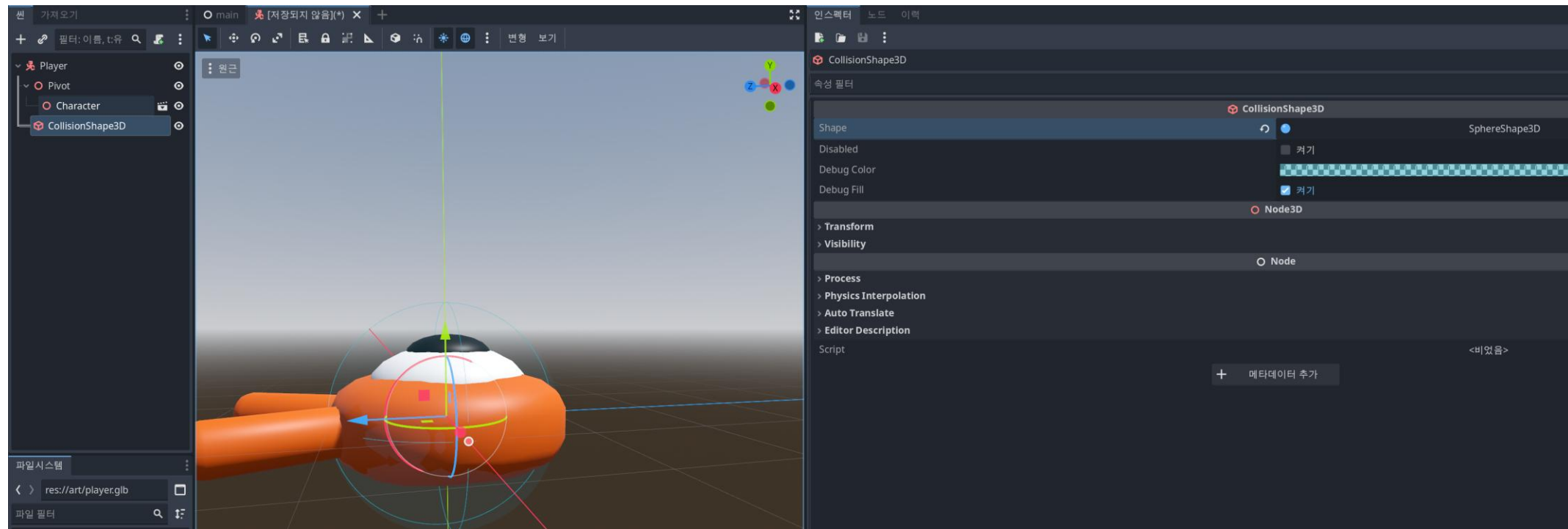


Pivot에 player.glb 올리기

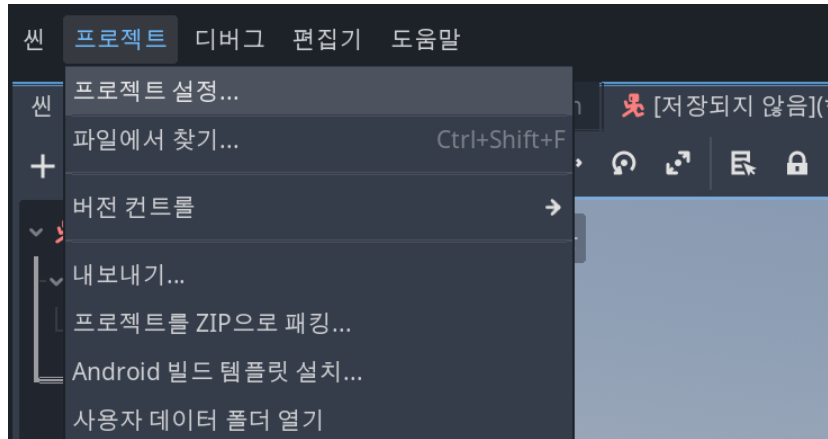
player 노드가 생기는데, 이를 Character로 변경



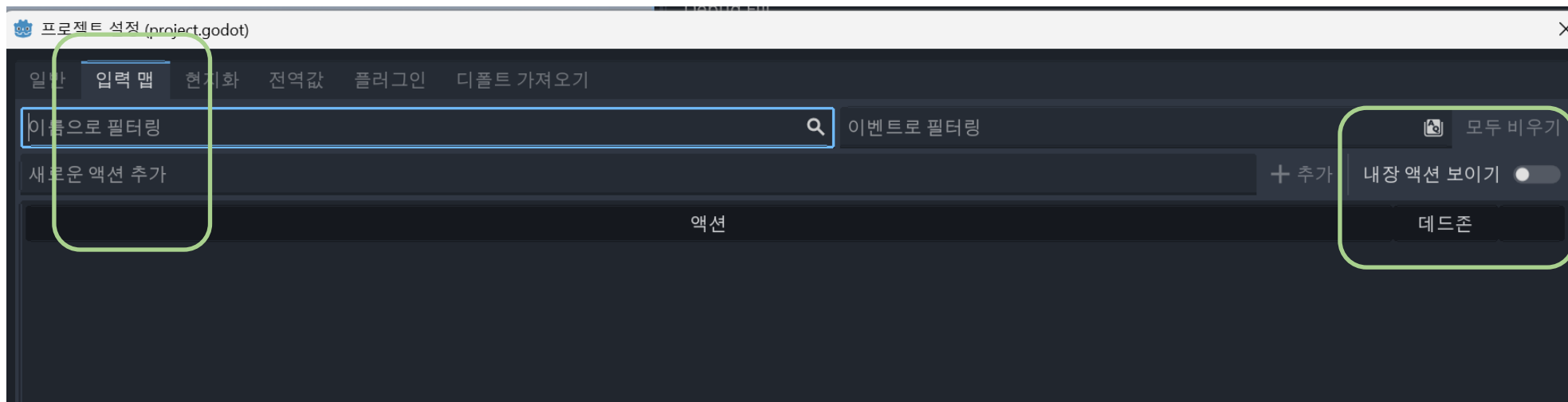
충돌 설정



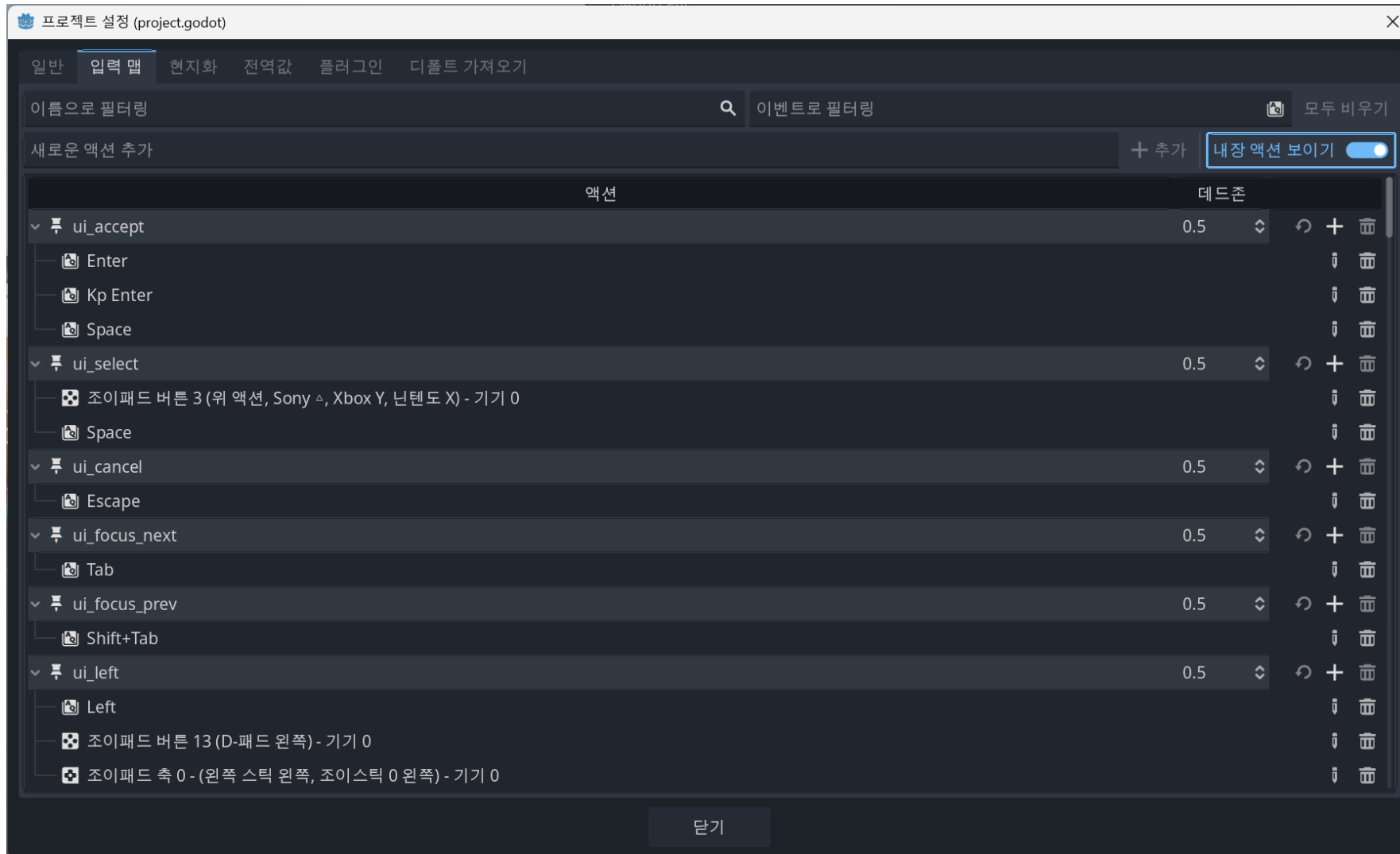
입력 처리



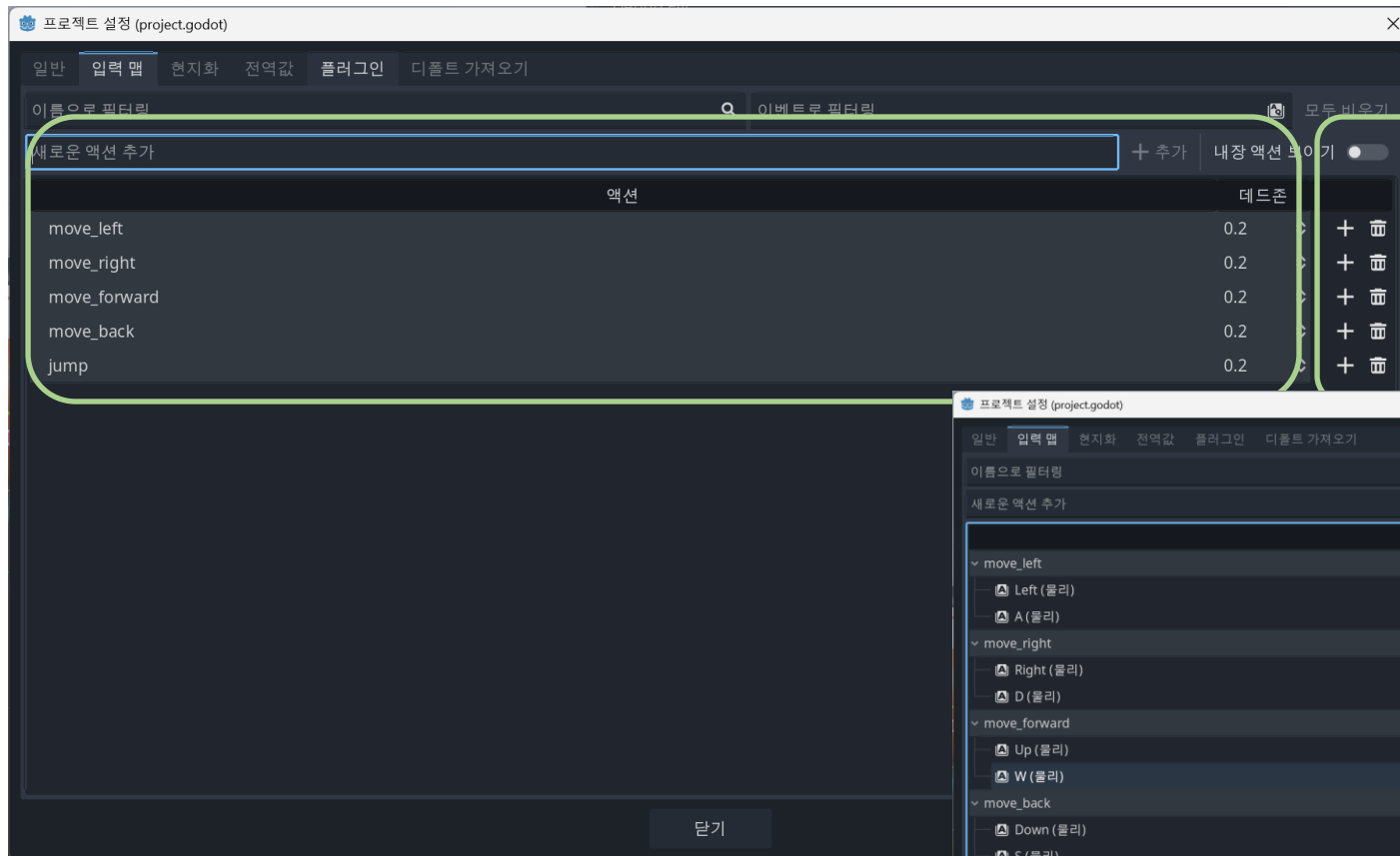
입력맵 → 내장 액션 보이기



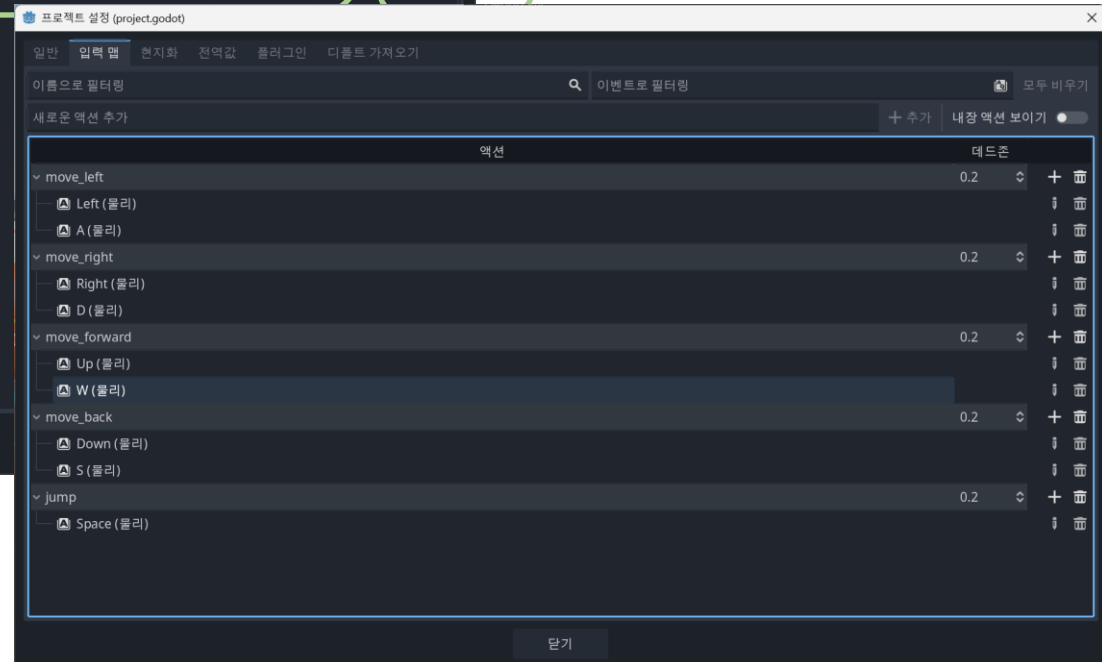
내장 액션



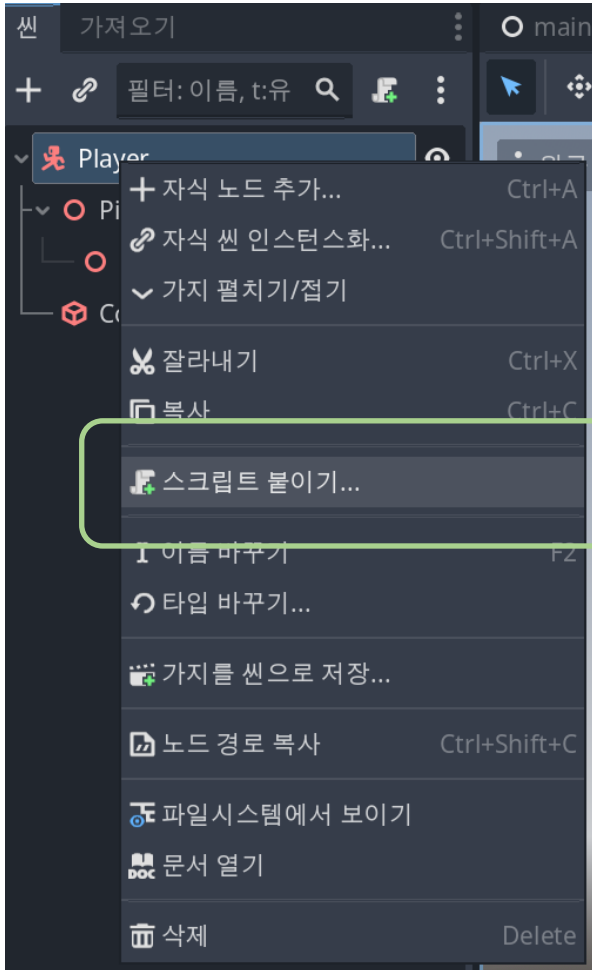
새로운 액션 추가



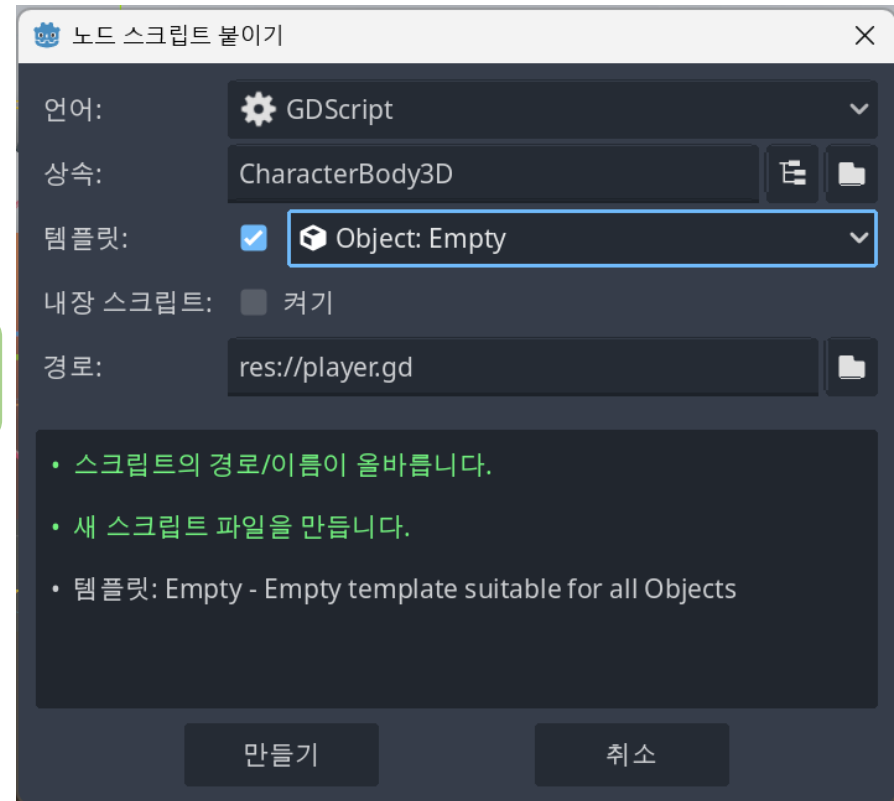
입력 추가: 키를 누르면 됨



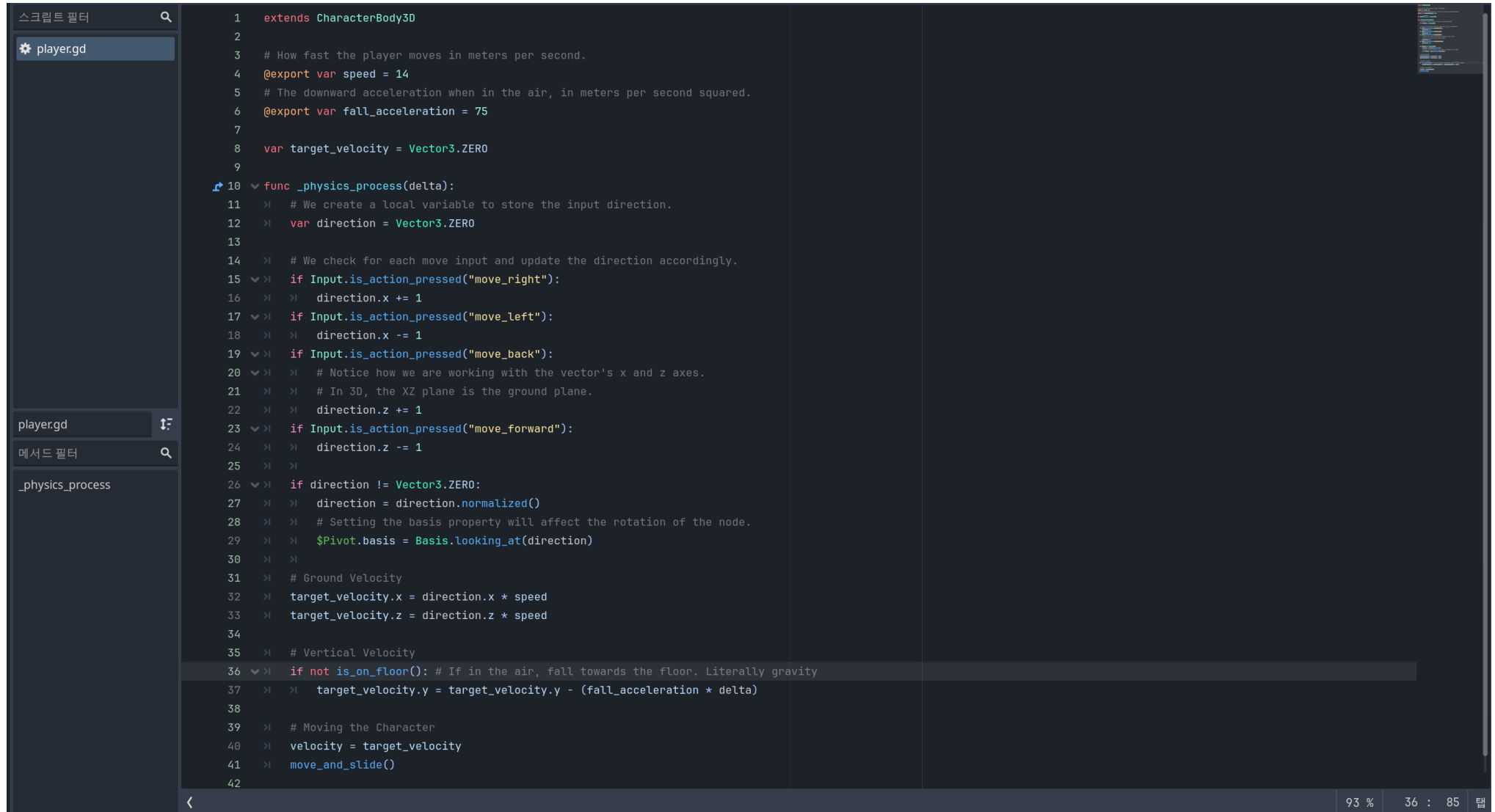
Time to Code



캐릭터에 스크립트 붙이기



코드 작성

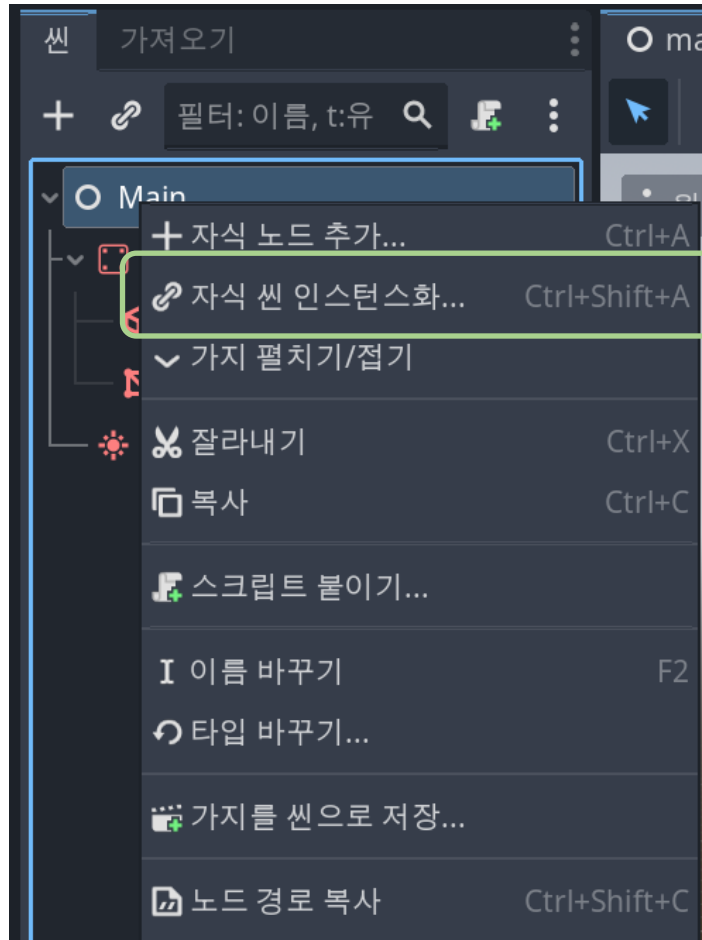


The screenshot displays the Godot Engine's code editor with the following components:

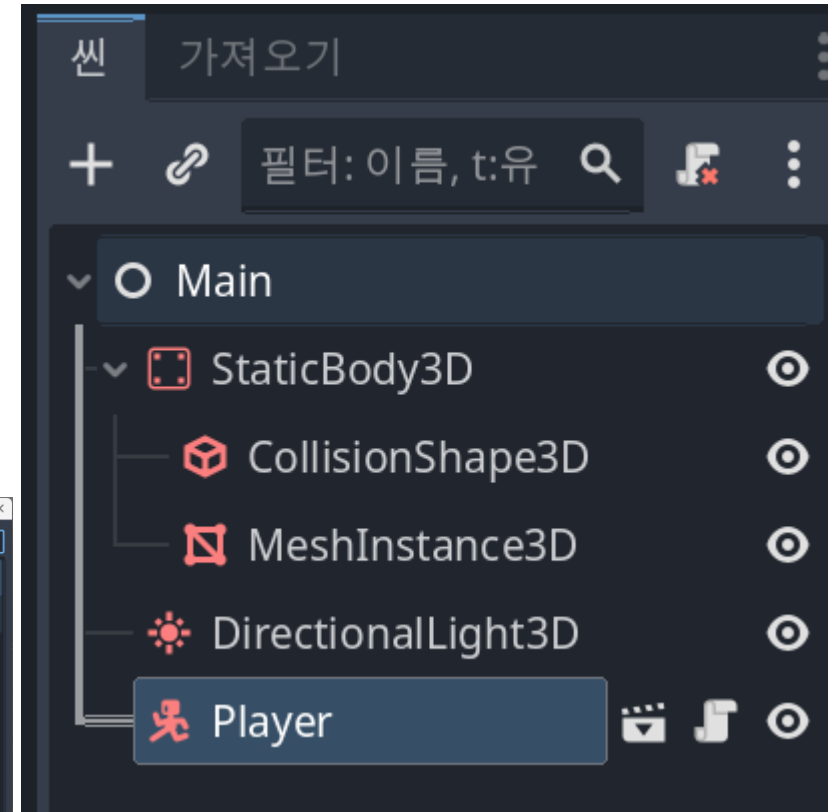
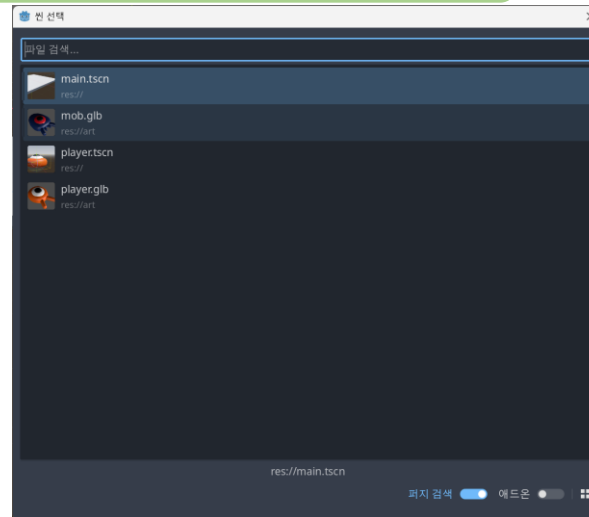
- Left Panel:** Contains a "스크립트 필터" (Script Filter) with a search icon. Below it, a list of scripts is shown, including "player.gd" (selected), "메서드 필터" (Method Filter), and "_physics_process".
- Main Editor:** Displays the code for the selected script, "player.gd". The code is written in GDScript and includes comments in Korean. The code defines a `CharacterBody3D` node, sets movement speed and fall acceleration, and implements the `_physics_process` function to handle movement and falling.
- Right Panel:** Shows a small preview of the game scene.
- Bottom Panel:** Displays the current zoom level (93 %) and the line numbers (36 : 85).

```
1  extends CharacterBody3D
2
3  # How fast the player moves in meters per second.
4  @export var speed = 14
5  # The downward acceleration when in the air, in meters per second squared.
6  @export var fall_acceleration = 75
7
8  var target_velocity = Vector3.ZERO
9
10 func _physics_process(delta):
11     # We create a local variable to store the input direction.
12     var direction = Vector3.ZERO
13
14     # We check for each move input and update the direction accordingly.
15     if Input.is_action_pressed("move_right"):
16         direction.x += 1
17     if Input.is_action_pressed("move_left"):
18         direction.x -= 1
19     if Input.is_action_pressed("move_back"):
20         # Notice how we are working with the vector's x and z axes.
21         # In 3D, the XZ plane is the ground plane.
22         direction.z += 1
23     if Input.is_action_pressed("move_forward"):
24         direction.z -= 1
25
26     if direction != Vector3.ZERO:
27         direction = direction.normalized()
28         # Setting the basis property will affect the rotation of the node.
29         $Pivot.basis = Basis.looking_at(direction)
30
31     # Ground Velocity
32     target_velocity.x = direction.x * speed
33     target_velocity.z = direction.z * speed
34
35     # Vertical Velocity
36     if not is_on_floor(): # If in the air, fall towards the floor. Literally gravity
37         target_velocity.y = target_velocity.y - (fall_acceleration * delta)
38
39     # Moving the Character
40     velocity = target_velocity
41     move_and_slide()
42
```

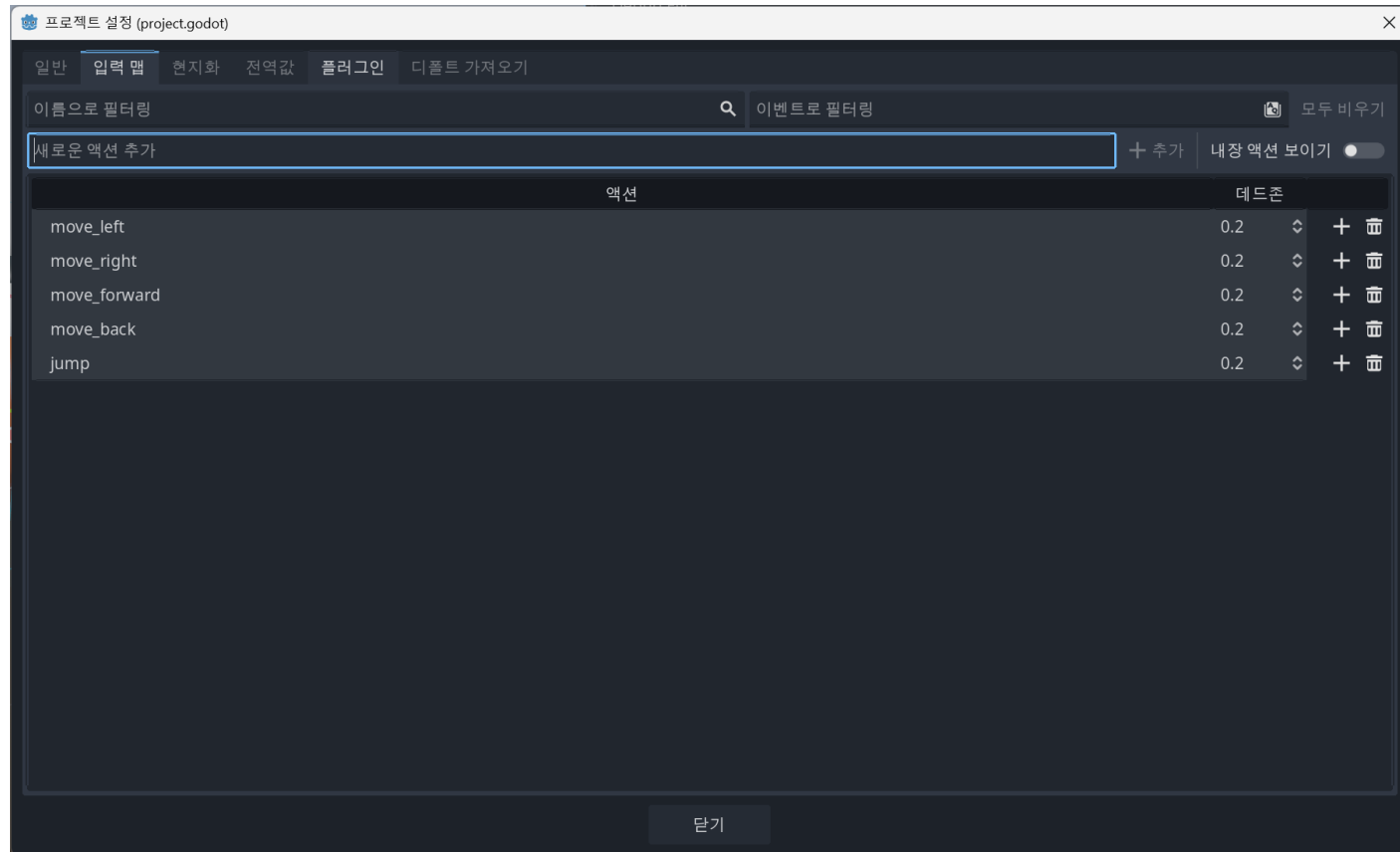
메인 씬에서 테스트 플레이어씬 저장하고 메인 씬을 다시 연다



자식 씬 인스턴스화



새로운 액션 추가

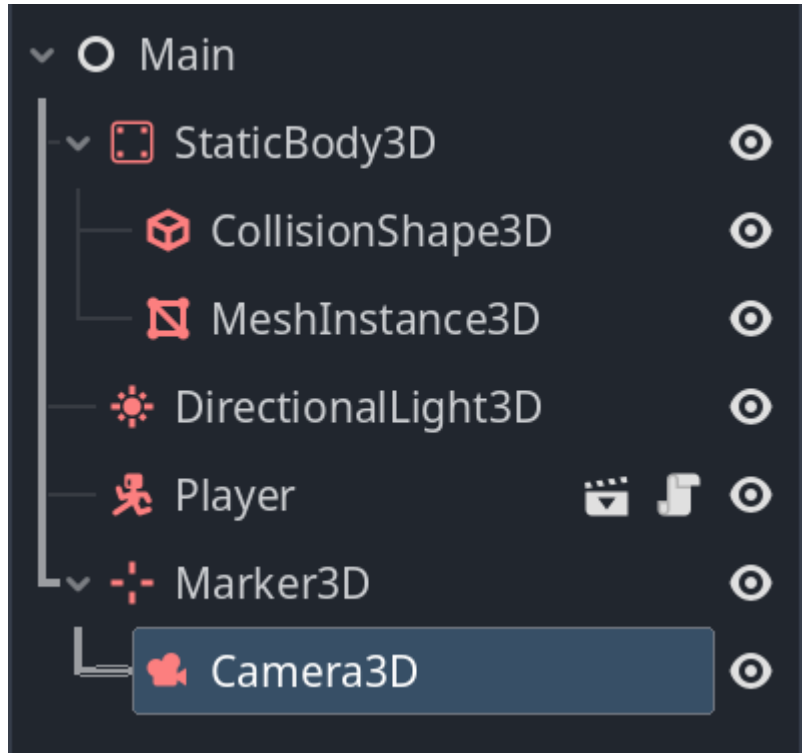


이 클래스가
우리의 캐릭터에
연결되어 있음

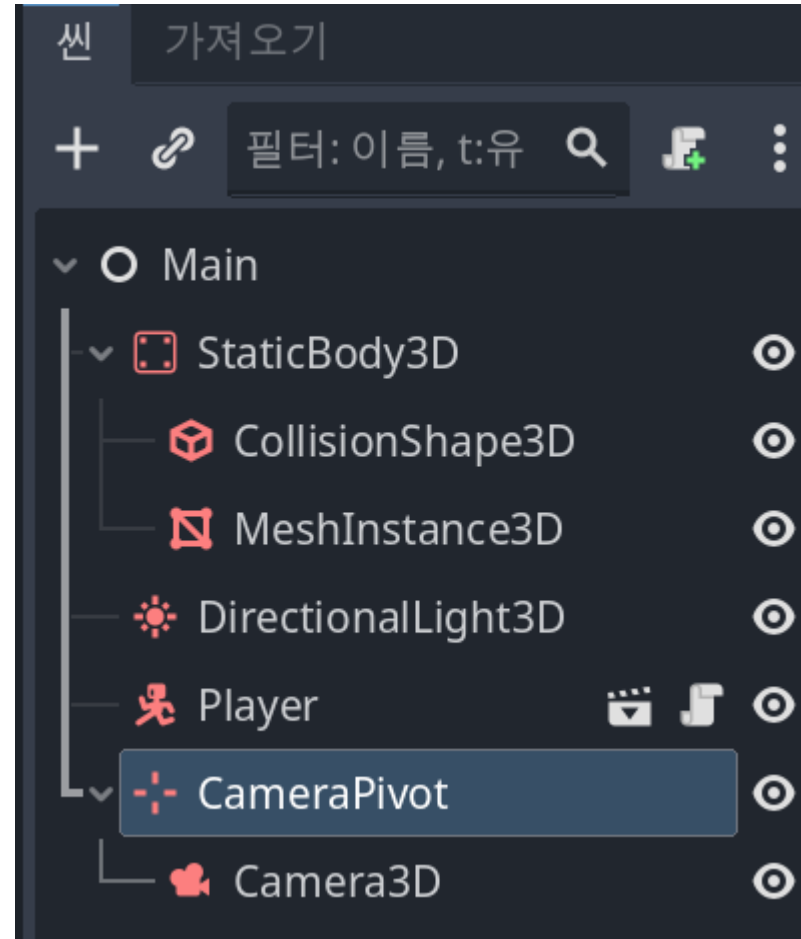
두 개의 클래스
Start, Update는
자동으로 호출되는
메소드

Start: 플레이될 때
Update: 매 프레임

카메라 추가



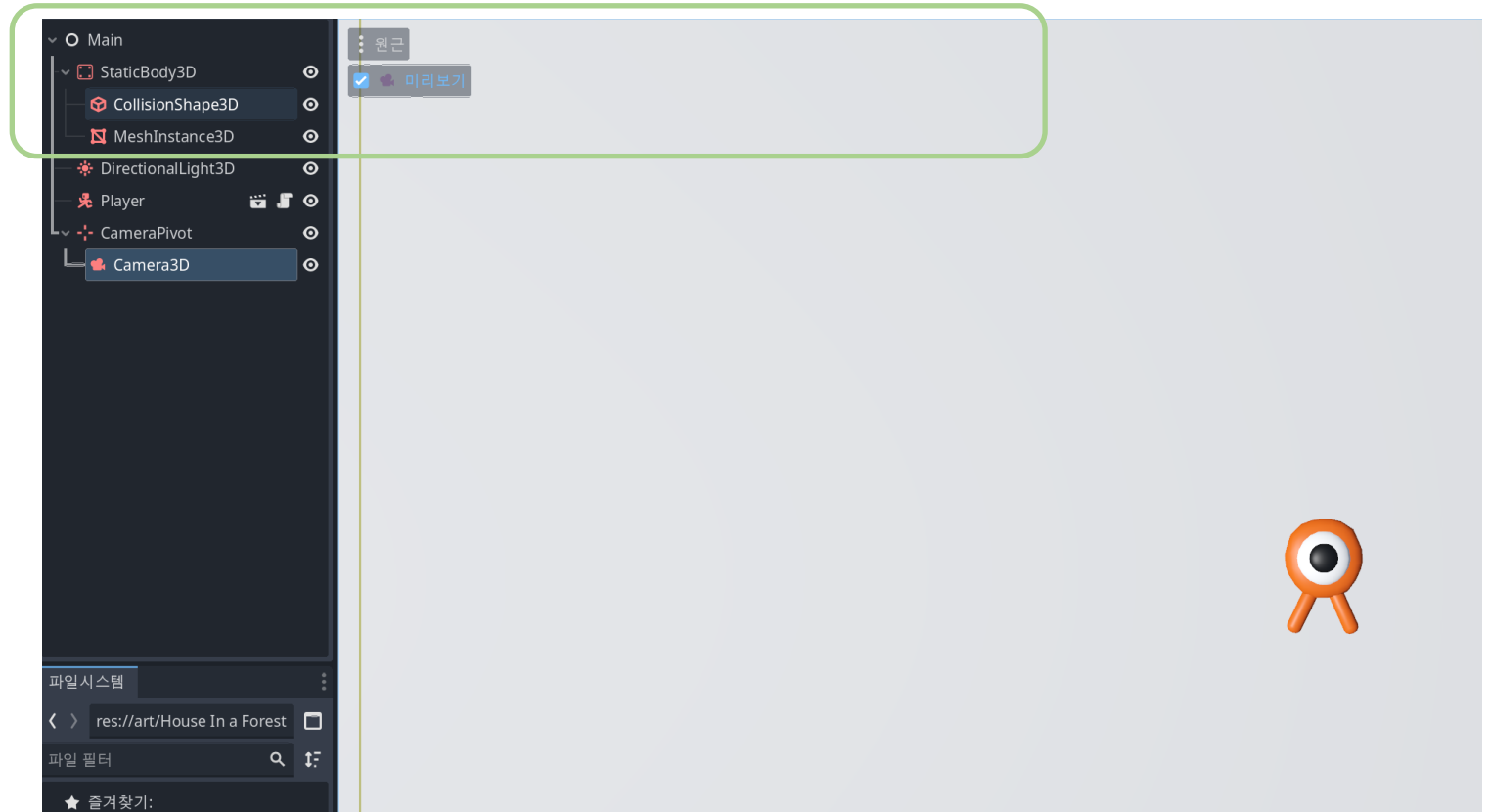
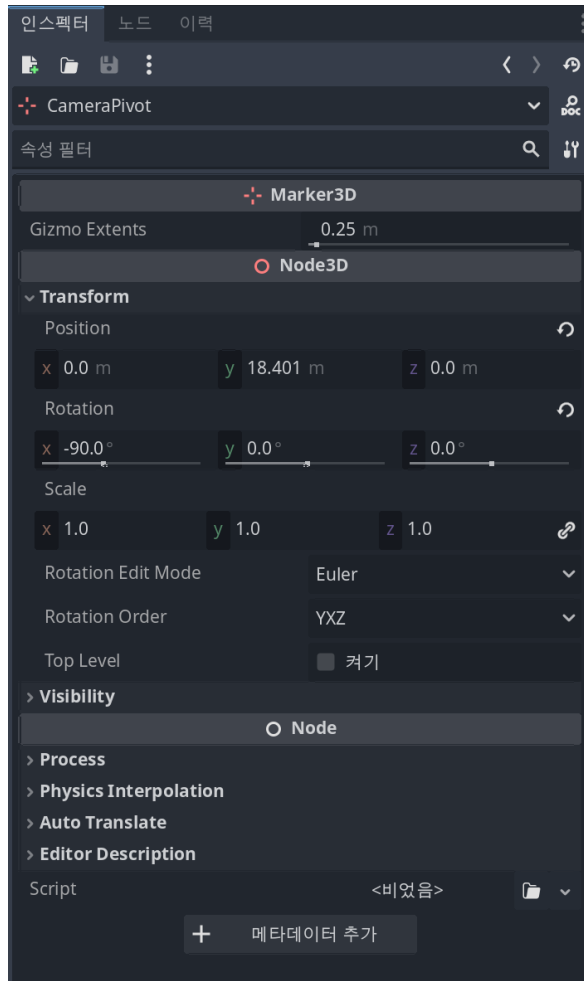
메인 씬에
Marker3D-Camera3D 달기



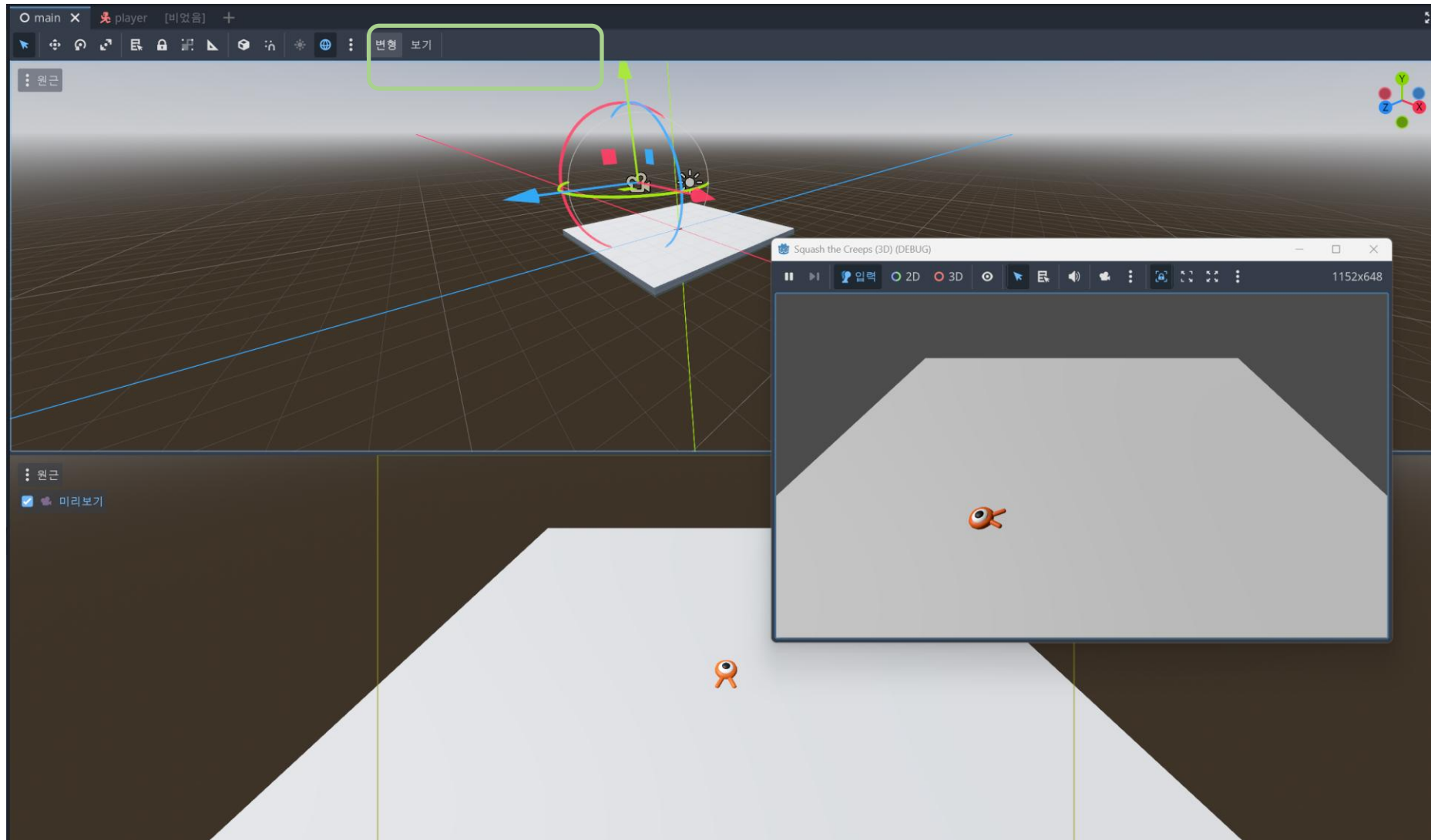
Marker3D를
CameraPivot으로
이름 변경

Pivot의 회전

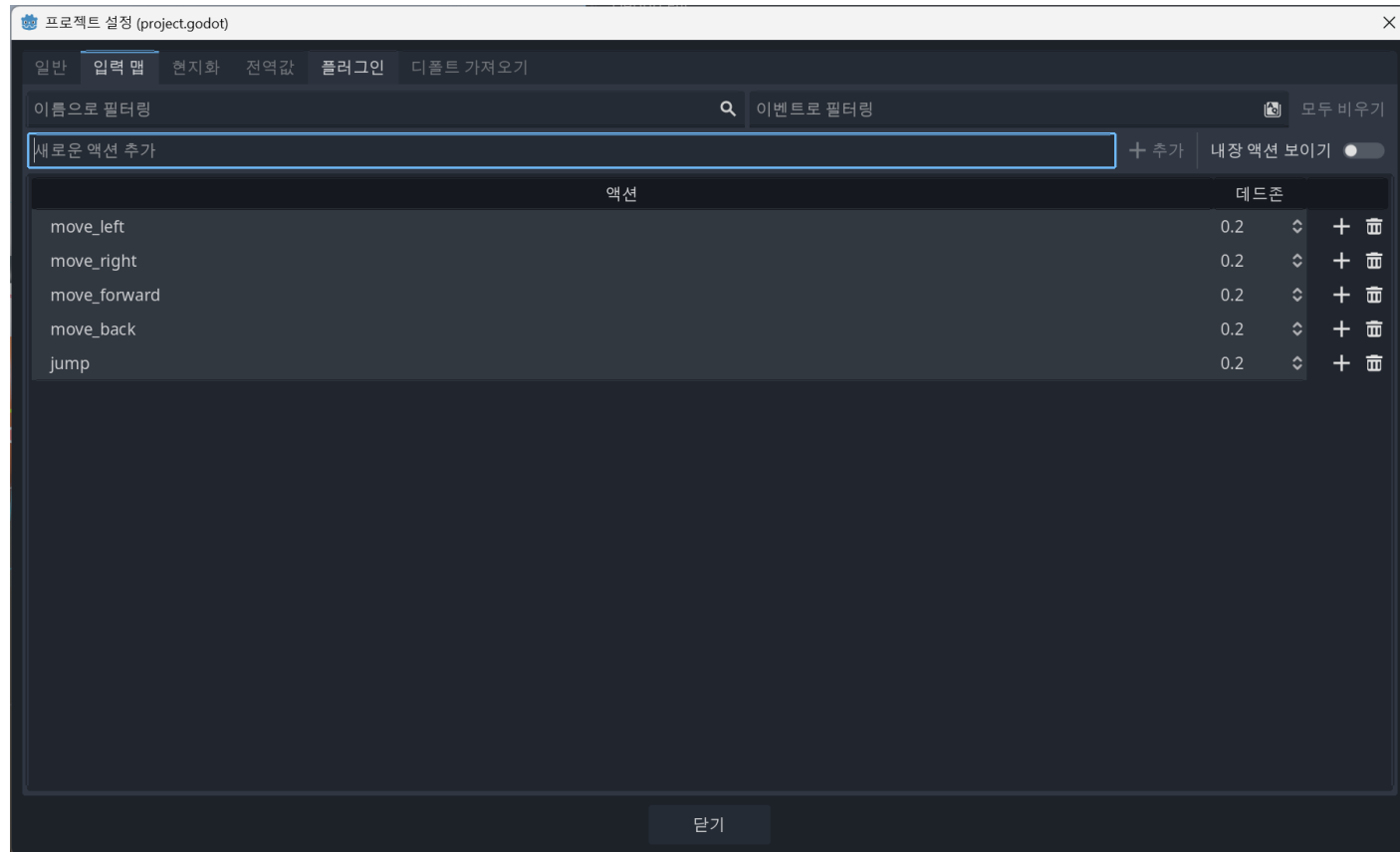
카메라 Preview



뷰포트 설정 및 게임 플레이 해보기



새로운 액션 추가



이 클래스가
우리의 캐릭터에
연결되어 있음

두 개의 클래스
Start, Update는
자동으로 호출되는
메소드

Start: 플레이될 때
Update: 매 프레임



Questions?